

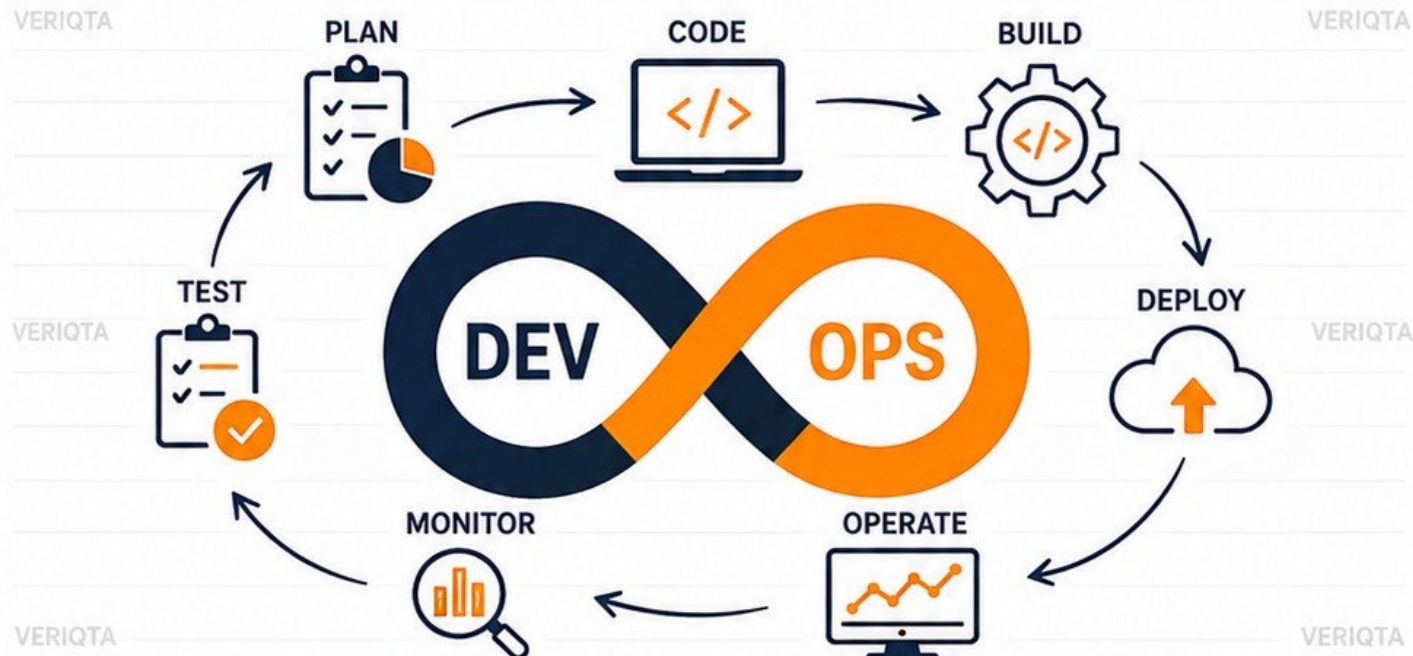


DEVOPS ENGINEER PROFESSIONAL

★ CERTIFICATION HANDBOOK ★

COMPLETE 40-PAGE MOCK EXAM HANDBOOK

✓ EXAM FOCUSED ✓ REALISTIC QUESTIONS ✓ DETAILED ANSWERS ✓ EXAM STRATEGY




EXAM-FOCUSED
CONTENT


200+ MOCK
QUESTIONS


DETAILED
EXPLANATIONS


PASS WITH
CONFIDENCE

AWS DEVOPS ENGINEER PROFESSIONAL CERTIFICATION HANDBOOK

⇒ HANDBOOK INTRODUCTION ⇒



THIS 40-PAGE HANDBOOK CONTAINS REALISTIC, SCENARIO-BASED MOCK EXAM QUESTIONS DESIGNED TO HELP YOU PASS THE **AWS CERTIFIED DEVOPS ENGINEER – PROFESSIONAL** EXAM.

1. HOW TO USE THIS HANDBOOK



- Work through one page (topic) at a time.
- Each page contains 5 scenario-based questions with detailed explanations.
- Attempt the questions before reading the answers.
- Understand why the correct answer is right and why others are wrong.
- Use it as a study, review, and confidence-building tool.

2. EXAM-STYLE QUESTION STRUCTURE



- Scenario-based, real-world situations.
- 4 answer choices (A, B, C, D).
- Only **ONE** best answer.
- Questions test your ability to design, deploy, operate and optimize.
- Focus on AWS architectural best practices, not service definitions.

3. SCENARIO-BASED REASONING



- Read the scenario carefully.
- Identify the real problem.
- Determine the **BEST** solution that meets all requirements.
- Think like a DevOps Engineer: reliability, scalability, security, automation, and cost.
- Always choose the most complete and operationally sound answer.

4. HOW TO ELIMINATE INCORRECT ANSWERS



- Eliminate answers that do **NOT** meet all requirements.
- Remove answers that are partially correct but incomplete.
- Avoid answers that introduce unnecessary complexity or cost.
- Eliminate answers that violate security, reliability, or best practices.
- Narrow down to the **BEST** option, not just a good option.

5. TIME-MANAGEMENT APPROACH



- You have 180 minutes for 75 questions.
- Average time per question: ~2.4 minutes.
- Keep moving—do not get stuck.
- Mark difficult questions and return later.
- Do not change answers unless you are confident.
- If unsure, make the best choice and move on.
- Review all marked questions in the final 30–40 minutes.
- Manage your time early to avoid rushing at the end.



TOTAL TIME
180 MINUTES

TARGET
~2.4 MIN / Q



SUCCESS IN THIS EXAM IS NOT ABOUT MEMORIZING SERVICES. IT IS ABOUT MAKING THE **BEST ARCHITECTURAL DECISIONS IN REAL-WORLD DEVOPS SCENARIOS.**



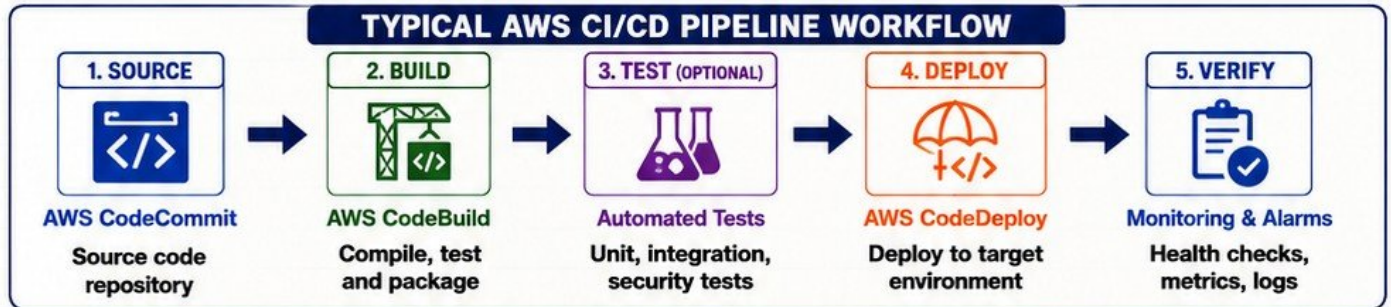
CI/CD PIPELINE ARCHITECTURE

CODEPIPELINE • CODEBUILD • CODEDEPLOY • WORKFLOWS • FAILURE ANALYSIS



CI/CD PIPELINE ARCHITECTURE: Automate building, testing and deploying applications reliably and securely using AWS native services.

TYPICAL AWS CI/CD PIPELINE WORKFLOW



5 SCENARIO QUESTIONS + ANSWERS

- 1** Your pipeline fails at the **BUILD** stage with exit code 137. What is the **MOST** likely cause?

A. IAM role does not have permissions
B. Build environment ran out of memory
C. CodeBuild project has wrong source type
D. Artifact store is encrypted

ANSWER B

Exit code 137 indicates the build container was killed, usually due to out-of-memory (OOM).
- 2** Deployments succeed but the application is not updated on EC2 instances. What should you check **FIRST**?

A. IAM permissions for CodeDeploy
B. CodeDeploy agent on the instances
C. Security group rules
D. ALB target health

ANSWER B

If CodeDeploy agent is missing or stopped, the deployment will not be executed on the instances.
- 3** You need artifacts created in CodeBuild to be available across multiple accounts. What should you use?

A. S3 bucket in each account
B. AWS CodeArtifact
C. Cross-account S3 artifact bucket
D. EFS file system

ANSWER C

Store artifacts in an S3 bucket in a shared account and grant cross-account access.
- 4** A pipeline is stuck in **APPROVAL** stage and no one received the approval request. Why?

A. EventBridge rule is misconfigured
B. SNS topic for approval notifications is incorrect
C. CodeBuild cache is corrupted
D. Pipeline artifact is too large

ANSWER B

Approval actions send notifications via SNS. If the SNS topic is wrong, no one receives the approval request.
- 5** After deploying a new version, alarms show high error rate and latency. What is the **BEST** immediate action?

A. Wait for auto scaling to fix it
B. Roll back the deployment
C. Increase instance size
D. Disable all alarms

ANSWER B

Rollback quickly restores the stable version and reduces impact.

KEY AWS SERVICES IN PIPELINE

SERVICE	ROLE
AWS CodePipeline	Orchestrates the entire CI/CD workflow.
AWS CodeBuild	Compiles code, runs tests and creates artifacts.
AWS CodeDeploy	Deploys application to EC2, On-Prem, Lambda or ECS.
Amazon S3	Stores source code artifacts, build artifacts and logs.
Amazon CloudWatch	Monitors pipeline, applications, logs and alarms.

PIPELINE FAILURE ANALYSIS CHECKLIST

- 1 Identify the failing stage (Source, Build, Deploy, etc.).
- 2 Check stage logs for the root error message.
- 3 Validate permissions (IAM roles and policies).
- 4 Verify network access (VPC, endpoints, security groups).
- 5 Check environment variables and secrets.
- 6 Validate artifact availability in S3.
- 7 Confirm target environment health.
- 8 Review CloudWatch metrics and alarms.
- 9 Re-run failed stage manually for detailed logs.
- 10 Fix the root cause and re-run the pipeline.

PRODUCTION BEST PRACTICES

- ✓ Use least privilege IAM roles.
- ✓ Always test in a separate staging environment.
- ✓ Enable CloudWatch alarms for all stages.
- ✓ Use blue/green or canary deployments.
- ✓ Store artifacts in encrypted S3 buckets.
- ✓ Automate rollbacks on failure.



PRODUCTION NOTE

A reliable CI/CD pipeline is not about building fast, but about deploying safely, detecting failures early, and recovering automatically.

AUTOMATE
EVERY STEP

+

VALIDATE
CONTINUOUSLY

+

MONITOR
EVERY CHANGE

+

ROLL BACK
FAST

=

RELIABLE
DELIVERY

DEPLOYMENT SUCCESS FORMULA

ADVANCED CODEPIPELINE

**MULTI-STAGE PIPELINES • CROSS-ACCOUNT DEPLOYMENTS • APPROVAL ACTIONS
ARTIFACT HANDLING • PIPELINE PERMISSIONS**



ADVANCED CODEPIPELINE: Build secure, scalable and maintainable delivery pipelines using best practices for enterprise environments.

MULTI-STAGE PIPELINE EXAMPLE



5 SCENARIO QUESTIONS + ANSWERS

1 You need to deploy to a different AWS account for production. What is the BEST way?

- A. Use a new CodePipeline in the target account
- B. Use cross-account role in CodeDeploy
- C. Use cross-account role in CodePipeline
- D. Share artifacts using public S3 bucket

ANSWER
C

CodePipeline can assume a cross-account IAM role to deploy into another account securely.

2 You want manual approval before deploying to production. What should you add?

- A. CloudWatch Alarm
- B. Lambda Approval Function
- C. Manual Approval action
- D. CodeBuild Test Report

ANSWER
C

Manual Approval action pauses the pipeline until a human approves or rejects the change.

3 Artifacts must be encrypted and shared across multiple accounts. What is the BEST approach?

- A. Public S3 bucket
- B. S3 bucket with SSE-KMS and bucket policy
- C. EFS file system
- D. CodeCommit repository

ANSWER
B

S3 with SSE-KMS and proper bucket policy ensures secure artifact storage and cross-account access.

4 Your pipeline fails with AccessDeniedException when CodePipeline tries to start CodeBuild. What should you check FIRST?

- A. CodeBuild compute type
- B. CodePipeline service role permissions
- C. Buildspec.yml file
- D. VPC endpoint configuration

ANSWER
B

CodePipeline needs permissions (IAM role) to start and manage CodeBuild.

5 You need to promote the SAME build artifact from Dev → QA → Prod without rebuilding. What should you do?

- A. Rebuild in each stage
- B. Use artifact from previous stage
- C. Store artifact in CodeCommit
- D. Use Lambda to copy files

ANSWER
B

Pass the output artifact from one stage to the next to ensure consistency and traceability.

ADVANCED CODEPIPELINE CAPABILITIES



MULTI-STAGE PIPELINES

Break down delivery into stages for better control, quality gates and visibility.



CROSS-ACCOUNT DEPLOYMENTS

Use IAM roles and trust relationships to deploy across accounts securely.



APPROVAL ACTIONS

Add Manual Approval or SNS Approval actions for governance and control.



ARTIFACT HANDLING

Store artifacts in S3. Pass artifacts between stages or across accounts.



PIPELINE PERMISSIONS

Use least-privilege IAM roles for CodePipeline, CodeBuild, CodeDeploy and S3.

PIPELINE FAILURE ANALYSIS CHECKLIST

- 1 Check the failing stage in CodePipeline execution details.
- 2 Review CloudWatch Logs for the failed action.
- 3 Verify IAM role permissions for the service.
- 4 Check artifact availability and S3 bucket access.
- 5 Validate environment variables and secrets.
- 6 Check network access (VPC, endpoints, security groups).
- 7 Verify service quotas and limits.
- 8 Re-run failed stage after fixing the root cause.

BEST PRACTICES

- ✓ Use least-privilege IAM roles.
- ✓ Enable encryption for all artifacts.
- ✓ Use cross-account roles for production deployments.
- ✓ Add approval for production changes.
- ✓ Always test in lower environments first.
- ✓ Use meaningful stage names.
- ✓ Enable CloudWatch logging for every stage.
- ✓ Automate rollbacks on failures.

SECURE CROSS-ACCOUNT DEPLOYMENT FLOW

CODEPIPELINE
(ACCOUNT A)



ASSUME ROLE
CROSS-ACCOUNT
IAM ROLE

DEPLOY
TO ACCOUNT B
(E.G., PROD)



VERIFY WITH
CLOUDWATCH



PRODUCTION NOTE

A well-designed pipeline is secure, observable, repeatable and easy to troubleshoot. Governance and automation are key to reliability.

AWS CODEBUILD

BUILDS, TESTS AND PACKAGES CODE AT SCALE



AWS CODEBUILD: A fully managed build service that compiles source code, runs tests, and produces software packages ready for deployment.

5 SCENARIO QUESTIONS + ANSWERS

- 1** Your CodeBuild build fails with "Exit code 137". What is the MOST likely cause?
- Permission denied to S3 bucket
 - Build environment ran out of memory
 - buildspec.yml syntax error
 - Network timeout to GitHub

ANSWER
B

Exit code 137 indicates the build container was killed, usually due to out-of-memory (OOM).

- 2** You need to pass a database password to CodeBuild securely. What is the BEST approach?
- Store in buildspec.yml file
 - Hardcode in environment variable
 - Use AWS Secrets Manager
 - Commit in Git and encrypt

ANSWER
C

Secrets Manager keeps sensitive data secure and can be referenced in CodeBuild environment variables.

- 3** Your build takes too long because dependencies are downloaded every time. What should you do?
- Increase compute type
 - Enable build caching
 - Disable source versioning
 - Run builds in parallel

ANSWER
B

Build caching stores dependencies between builds, significantly reducing build time.

- 4** CodeBuild cannot access a private GitHub repository. What is the MOST likely issue?
- Missing VPC endpoint
 - Incorrect IAM permissions
 - Wrong build image
 - CloudWatch Logs issue

ANSWER
B

CodeBuild needs IAM permissions (or Git credentials) to access private repositories.

- 5** A build fails during the INSTALL phase. What should you check FIRST?
- CloudWatch alarms
 - Build logs for dependency errors
 - CodePipeline configuration
 - S3 artifact bucket

ANSWER
B

Install phase failures are usually due to dependency or package installation errors. Check build logs first.

KEY CODEBUILD CONCEPTS



BUILD PROJECT
Defines source, environment, build spec, and artifacts.



BUILD ENVIRONMENT
Compute type, image, privileged mode, network settings.



BUILD SPEC
buildspec.yml controls phases, commands and artifacts.



ARTIFACTS
Output files stored in Amazon S3 or passed to next stage.



ENV VARIABLES
Store configuration values for builds securely.



SECRETS
Use Secrets Manager or Parameter Store for sensitive data.



BUILD CACHING
Cache dependencies to speed up subsequent builds.



LOGGING
Logs sent to CloudWatch Logs for debugging.



PERMISSIONS
IAM role controls access to AWS resources.

BUILD SPEC (buildspec.yml) EXAMPLE

```
version: 0.2
phases:
  install:
    runtime-versions:
      nodejs: 18
    commands:
      - echo Installing dependencies...
      - npm ci
  pre_build:
    commands:
      - echo Running tests...
      - npm test
  build:
    commands:
      - echo Building application...
      - npm run build
artifacts:
  files:
    - '**/*'
  discard-paths: yes
```

ENVIRONMENT VARIABLES

- Plain text variables for non-sensitive data.
- Can be used in buildspec.yml as: \$VAR_NAME
- Override at build time if needed.

SECRETS

- Store secrets in Secrets Manager or Parameter Store.
- Reference in CodeBuild:
 - name: DB_PASSWORD
 - type: SECRETS_MANAGER
 - value: arn:aws:secretsmanager:region:account-id:secret:mysecret
- Never hardcode secrets.

BUILD CACHING

- Enable in project settings.
- Supports LOCAL, S3, or CUSTOM cache.
- Common use: npm, pip, Maven, Gradle, Docker layers.

COMMON CACHE LOCATIONS

Node.js	~/.npm
Python	~/.cache/pip
Maven	~/.m2/repository
Gradle	~/.gradle/caches

FAILED BUILD TROUBLESHOOTING GUIDE

SYMPTOM	POSSIBLE CAUSE	WHAT TO CHECK
Build fails with Exit code 137	Out of memory (OOM)	- Increase compute type - Optimize build steps - Reduce memory usage
Permission denied errors	IAM role lacks permissions	- Check IAM role policies - Verify S3, ECR, Secrets Manager access
Cannot access Git repository	Missing permissions or Git credentials	- Check IAM role permissions - Verify SSH/HTTPS credentials
Install phase fails	Dependency or package installation error	- Check build logs - Verify package versions
Build is slow	No caching or large dependencies	- Enable build caching - Use smaller base image

CODEBUILD BEST PRACTICES

- Use least-privilege IAM roles.
- Store secrets securely.
- Enable build caching.
- Use smaller, purpose-built images.
- Keep build logs organized.
- Fail fast and validate early.
- Automate everything.
- Monitor with CloudWatch.

VIEW AND ANALYZE BUILD LOGS

1. IN CODEBUILD CONSOLE

- Open the build
- Click "View logs"
- Examine each phase



CLOUDWATCH LOGS

2. USING CLI

```
aws logs describe-log-streams \
  --log-group-name /aws/codebuild/<project>
aws logs get-log-events \
  --log-group-name /aws/codebuild/<project> \
  --log-stream-name <stream>
```

Look for **ERROR**, **WARNING** and stack traces to identify the root cause quickly.



PRODUCTION NOTE

A reliable build pipeline is the foundation of a strong DevOps practice. Secure your pipeline, optimize build performance, and make failures easy to detect and fix.

AWS CODEDEPLOY

AUTOMATE DEPLOYMENTS TO ANYWHERE WITH CONFIDENCE



AWS CODEDEPLOY: Automates application deployments to EC2, On-Premises and ECS with zero downtime and built-in rollback capabilities.

5 SCENARIO QUESTIONS + ANSWERS

- You need to deploy a new version to EC2 instances with zero downtime and automatic rollback. What is the BEST deployment type?

A. In-place deployment
B. Blue/Green deployment
C. All-at-once deployment
D. Rolling deployment

ANSWER B

Blue/Green deployment Route traffic to a healthy new environment and allows automatic rollback.
- You have an application on EC2 and want the simplest deployment method. What should you use?

A. Blue/Green deployment
B. In-place deployment
C. Canary deployment
D. External deployment

ANSWER B

In-place deployment installs the new version directly on the existing instances.
- Which component in CodeDeploy defines the set of instances (or targets) where the application is deployed?

A. Application
B. Deployment Group
C. Revision
D. Lifecycle Hook

ANSWER B

A Deployment Group contains the target instances, deployment config, and other settings.
- You need to run custom scripts before and after the deployment on EC2 instances. What should you configure?

A. CloudWatch Events
B. Lifecycle hooks
C. CodeBuild phases
D. Deployment alarms

ANSWER B

Lifecycle hooks (BeforeInstall, AfterInstall, ApplicationStart, ValidateService, etc.) allow custom actions.
- A deployment failed after traffic was shifted. What is the BEST action to restore the previous working version?

A. Manually reinstall the previous version
B. Use CodeDeploy rollback
C. Stop the instances
D. Delete the deployment group

ANSWER B

CodeDeploy automatically rolls back to the last known good revision in a Blue/Green or In-Place deployment.

CODEDEPLOY TROUBLESHOOTING GUIDE

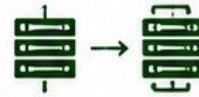
ISSUE	POSSIBLE CAUSE	WHAT TO CHECK
Deployment fails	Scripts or lifecycle hook error	Check lifecycle logs in <code>/var/log/aws/codedeploy-agent/</code>
Permission denied	IAM role lacks permissions	Verify CodeDeploy service role permissions
Revision not found	S3 bucket or key incorrect	Check bucket policy, object path and region
Health check failed	Application not healthy	Check target group health, cloudwatch alarms, logs
Deployment timeout	Hooks or scripts take too long	Increase timeout settings or optimize scripts

PRODUCTION NOTE

CodeDeploy is most powerful when combined with CodePipeline, CloudWatch, IAM and S3. Design for safety first, then speed.

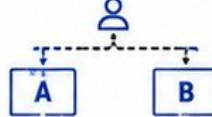
DEPLOYMENT TYPES IN CODEDEPLOY

IN-PLACE DEPLOYMENT



- New version installed on existing instances.
- Simple and fast.
- Short downtime possible.
- Rollback by reinstalling previous version.

BLUE/GREEN DEPLOYMENT



- New version deployed to a new environment.
- Traffic shifted after validation.
- Zero downtime.
- Automatic rollback by routing back.

CANARY DEPLOYMENT



- New version deployed to a small subset.
- Monitor before full rollout.
- Lower risk.
- Gradual traffic shifting.

CODEDEPLOY DEPLOYMENT WORKFLOW (EC2 EXAMPLE)



LIFECYCLE HOOKS (EC2)

- BeforeInstall – Prepare for installation.
- AfterInstall – Run after deployment files are installed.
- ApplicationStart – Start or restart application.
- ValidateService – Validate the application is healthy.
- BeforeBlockTraffic – Before traffic is routed (Blue/Green).
- AfterAllowTraffic – After traffic is routed (Blue/Green).

ROLLBACK BEHAVIOR

- Automatic rollback in Blue/Green when alarms fail.
- Automatic rollback in In-Place when health checks fail.
- Manual rollback can be triggered at any time.
- CodeDeploy restores the last known good revision.
- Rollback is logged and visible in deployment history.

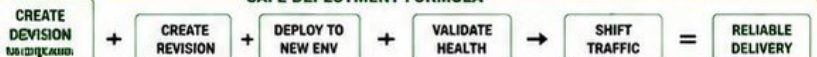
KEY CODEDEPLOY COMPONENTS

COMPONENT	DESCRIPTION
APPLICATION	Logical grouping of your deployable resources.
DEPLOYMENT GROUP	Defines where and how to deploy (targets, config, roles).
REVISION	The version of your application (bundle, zip, image, etc.).
DEPLOYMENT CONFIG	Defines deployment type and success/rollback settings.
SERVICE ROLE	IAM role CodeDeploy uses to perform actions.

CODEDEPLOY BEST PRACTICES

- ✓ Use Blue/Green for critical applications.
- ✓ Enable CloudWatch alarms for automatic rollback.
- ✓ Use meaningful deployment group names.
- ✓ Store deployment artifacts in encrypted S3.
- ✓ Keep lifecycle hooks fast and reliable.
- ✓ Validate before shifting traffic.
- ✓ Automate deployments through CodePipeline.
- ✓ Monitor deployments with CloudWatch.

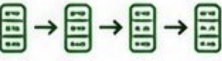
SAFE DEPLOYMENT FORMULA



AWS DEVOPS ENGINEER PROFESSIONAL DEPLOYMENT STRATEGIES

★ 5 SCENARIO QUESTIONS + ANSWERS ★

DEPLOYMENT STRATEGY OVERVIEW

ROLLING DEPLOYMENT	BLUE/GREEN DEPLOYMENT	CANARY DEPLOYMENT	LINEAR DEPLOYMENT	ALL-AT-ONCE DEPLOYMENT
 - Replace instances in batches - Some old + new run together - Lower risk - Takes more time	 - Two identical envs (Blue & Green) - Switch traffic at once - Very safe - Higher cost	 - Send traffic to new version (small %) - Monitor metrics - Gradually increase - Very safe + flexible	 - Increase traffic in fixed steps - Monitor between steps - Balanced risk and speed	 - Deploy new version to all at once - Fastest - Highest risk - Hard to rollback

5 SCENARIO QUESTIONS + ANSWERS

1  A company wants to minimize downtime and ensure instant rollback if issues occur after deployment. Which strategy is MOST suitable?	A. Rolling Deployment B. Blue/Green Deployment C. All-at-Once Deployment D. Linear Deployment	ANSWER: B Blue/Green allows instant traffic switch back to the old environment with zero downtime if problems occur.
2  You want to release a new version to a small subset of users, monitor metrics, and then increase traffic gradually. Which strategy should you use?	A. Canary Deployment B. Rolling Deployment C. All-at-Once Deployment D. Blue/Green Deployment	ANSWER: A Canary sends a small % of traffic to the new version first, allowing monitoring before expanding.
3  Your application has multiple instances behind an ALB. You need to deploy with minimal risk and keep the service available. Which strategy is BEST?	A. Rolling Deployment B. All-at-Once Deployment C. Blue/Green Deployment D. Linear Deployment	ANSWER: A Rolling replaces instances in batches, ensuring some healthy instances are always available.
4  You have limited infrastructure budget but need safer deployments than all-at-once. Which strategy provides a good balance?	A. Linear Deployment B. Blue/Green Deployment C. All-at-Once Deployment D. Canary Deployment	ANSWER: A Linear increases traffic in steps (10%, 20%, 50%, 100%), balancing risk, speed, and cost.
5  A critical financial system must avoid bugs reaching all users. Which strategy offers the SAFEST deployment?	A. All-at-Once Deployment B. Blue/Green Deployment C. Rolling Deployment D. All options are equally safe	ANSWER: B Blue/Green is the safest because the new version is fully tested in isolation before any user sees it.

CHOOSING THE SAFEST STRATEGY



• HIGHEST SAFETY:

• LOWEST DOWNTIME:

• LOWEST COST:

• FASTEST DEPLOYMENT:

BLUE/GREEN > CANARY > LINEAR > ROLLING > ALL-AT-ONCE

BLUE/GREEN > CANARY > LINEAR > ROLLING > ALL-AT-ONCE

ROLLING > LINEAR > CANARY > BLUE/GREEN > ALL-AT-ONCE

ALL-AT-ONCE > BLUE/GREEN > ROLLING > LINEAR > CANARY



KEY TAKEAWAY

THERE IS NO ONE-SIZE-FITS-ALL STRATEGY. CHOOSE BASED ON RISK TOLERANCE, DOWNTIME ACCEPTANCE, COST, AND RELEASE VELOCITY.



EXAM TIP

LOOK FOR KEYWORDS LIKE "MINIMIZE DOWNTIME", "INSTANT ROLLBACK", "SMALL SUBSET", "GRADUALLY INCREASE", OR "LOW COST" TO PICK THE BEST ANSWER.

AWS DEVOPS ENGINEER PROFESSIONAL INFRASTRUCTURE AS CODE

★ 5 SCENARIO QUESTIONS + ANSWERS ★

TOPIC OVERVIEW

CLOUDFORMATION	NESTED STACKS	CHANGE SETS	STACKSETS	DRIFT DETECTION	FAILED STACK RECOVERY
 • Declarative IaC • Stacks • Templates (YAML/JSON) • Resources • Outputs	 • Modular templates • Reuse components • Better organization • Easier updates	 • Preview changes • See what will be modified • Reduce risk • Requires review	 • Deploy stacks across accounts & regions • Multi-account management • Consistent governance	 • Detect manual changes • Compare actual vs template • Maintain compliance	 • Diagnose failure • Fix template/resources • Recover safely

5 SCENARIO QUESTIONS + ANSWERS

1 	You need to preview the changes before updating a production stack to ensure no unintended resources will be modified or deleted. What should you use?	A. Drift Detection B. Change Set C. Stack Policy D. Termination Protection	ANSWER: B Change Sets allow you to preview what changes CloudFormation will make before you execute the update.
2 	Your organization wants to deploy the same CloudFormation template to multiple AWS accounts and regions consistently. Which service is designed for this?	A. Nested Stacks B. StackSets C. CloudFormation Macros D. Change Sets	ANSWER: B StackSets enables you to deploy stacks across multiple accounts and regions from a central location.
3 	You have a large CloudFormation template that is becoming hard to maintain. You want to break it into smaller, reusable components. What should you use?	A. Nested Stacks B. StackSets C. AWS Service Catalog D. Change Sets	ANSWER: A Nested Stacks let you split a large template into smaller templates and reuse them.
4 	A resource in your stack was manually modified in the AWS Console. You need to identify which resources are different from the template. What should you use?	A. CloudWatch Events B. Drift Detection C. Stack Export D. Resource Import	ANSWER: B Drift Detection compares actual resource configuration with your template and identifies differences.
5 	A CloudFormation stack update failed and is now in ROLLBACK_FAILED state. You need to recover the stack so you can continue using it. What is the BEST approach?	A. Delete the stack and recreate it B. Fix the issue and continue rollback C. Retain failed resources, fix the issue, and update the stack D. Disable rollback and retry the update	ANSWER: C Retain the failed resources (using Continue Update Rollback or resources-to-retain), fix the issue, then update the stack.

BEST PRACTICES



- ✓ Always use **CHANGE SETS** for production updates.
- ✓ Enable **TERMINATION PROTECTION** on critical stacks.
- ✓ Use **NESTED STACKS** to keep templates modular.
- ✓ Use **STACKSETS** for multi-account and multi-region deployments.
- ✓ Run **DRIFT DETECTION** regularly to ensure compliance.
- ✓ Have a documented recovery plan for failed stacks.



PRO TIP

Practice reading Change Set outputs. Look for resources to be deleted, replaced, or modified. This skill is heavily tested in the exam.

AWS DEVOPS ENGINEER PROFESSIONAL CLOUDFORMATION ADVANCED SCENARIOS

★ 5 SCENARIO QUESTIONS + ANSWERS ★

TOPIC OVERVIEW

CROSS-STACK REFERENCES	UPDATE POLICIES	CREATION POLICIES	DELETION POLICIES	STACK DEPENDENCIES	ROLLBACK FAILURES
					
- Share outputs between stacks - Export / ImportValue - Loose coupling - Avoid circular dependencies	- Control how resources update - Auto Scaling, RDS, ECS - Minimize downtime - Preserve stability	- Ensure resources are ready - Wait for signals - Prevents stack from completing too early	- Retain or delete on stack delete - Retain important data - Prevent accidental loss	- Control order of stack creation - DependsOn - Avoid race conditions - Ensure reliability	- Rollback can fail - Resources stuck in DELETE_FAILED - Manual recovery required

5 SCENARIO QUESTIONS + ANSWERS

1		You have two CloudFormation stacks: Network Stack and Application Stack. The Application Stack needs the VPC ID created in the Network Stack. What is the BEST way to achieve this?	A. Hardcode the VPC ID B. Export the VPC ID from the Network Stack and Import it in the Application Stack C. Use a Parameter Store value D. Create the VPC in both stacks	ANSWER: B Using Outputs + Export in one stack and ImportValue in another is the best way to reference values across stacks while maintaining loose coupling.
2		You need to update an Auto Scaling Group in a CloudFormation stack. You want to ensure the new instances are healthy before the old ones are terminated. What should you use?	A. Creation Policy B. Update Policy C. Deletion Policy D. Stack Policy	ANSWER: B Update Policy for Auto Scaling allows you to control rolling updates and ensure minimum healthy capacity during updates.
3		Your EC2 instances take several minutes to configure after launch. You want CloudFormation to wait until the instances are fully ready before continuing. What should you use?	A. Creation Policy with Resource Signals B. Update Policy C. Deletion Policy D. DependsOn	ANSWER: A Creation Policy with Resource Signals ensures CloudFormation waits for the instances to signal success before completing the stack creation.
4		You have an RDS database in a CloudFormation stack. If the stack is deleted, you want to keep the database. What should you configure?	A. Update Policy B. Deletion Policy: Retain C. Creation Policy D. DependsOn	ANSWER: B Deletion Policy: Retain ensures the resource is not deleted when the stack is deleted.
5		You have three stacks: Network Stack, Security Stack, and Application Stack. The Application Stack requires resources from both the Network and Security stacks. How should you manage the deployment order?	A. Use Stack Dependencies (DependsOn) B. Use Parameters C. Use Update Policy D. Use Deletion Policy	ANSWER: A Stack Dependencies (DependsOn) ensures stacks are created in the correct order based on their requirements.
		ROLLBACK FAILURES SCENARIO A stack update failed and is now in ROLLBACK_FAILED state. Some resources are stuck in DELETE_FAILED. What is the BEST approach to recover the stack?	A. Delete the stack and recreate it B. Fix the failed resources, continue the rollback, then update the stack C. Disable rollback and retry the update D. Ignore and create a new stack	ANSWER: B Fix the failed resources first (manually or via CLI), continue the rollback or use Continue Update Rollback, then update the stack.

BEST PRACTICES



- Use EXPORTED OUTPUTS for cross-stack references.
- Apply UPDATE POLICIES to critical services for zero/minimal downtime.
- Use CREATION POLICIES for instances that need initialization.
- Set DELETION POLICIES to retain important data.
- Define STACK DEPENDENCIES to ensure correct deployment order.
- Document RECOVERY PROCEDURES for rollback failure scenarios.



PRO TIP

Always test stack updates in a non-production account first. Review Change Sets carefully, use least-privilege IAM roles, and have a rollback plan before making changes in production.

AWS DEVOPS ENGINEER PROFESSIONAL

AWS CDK AND INFRASTRUCTURE AUTOMATION

★ 5 SCENARIO QUESTIONS + ANSWERS ★

TOPIC OVERVIEW				
CDK STACKS  - Collection of AWS resources - Maps to a single CloudFormation stack - Define scope and synth output	CONSTRUCTS  - Reusable components - L1 (CFN), L2 (AWS services), L3 (patterns) - Promote reusability and consistency	ENVIRONMENTS  - Account + Region configuration - Multi-env support (dev, test, prod) - Environment-aware deployments	DEPLOYMENT WORKFLOWS  - Synth → Diff → Deploy - Bootstrapping (CDK Toolkit) - CI/CD integration - Approval and promotion across stages	CLOUDFORMATION INTEGRATION  - CDK synthesizes to CloudFormation - Leverages all CFN capabilities - Outputs, Exports, Parameters

5 SCENARIO QUESTIONS + ANSWERS

1 	You need to deploy a set of AWS resources as a single unit that can be updated and rolled back together. Which CDK concept should you use?	A. Construct B. Stack C. Environment D. Pipeline	ANSWER: B A CDK Stack represents a collection of resources and maps to a single CloudFormation stack.
2 	You want to reuse a VPC with common subnets, routing, and security settings across multiple CDK projects. What should you build?	A. L1 Construct B. L2 Construct C. L3 Construct D. CDK Stack	ANSWER: C L3 Constructs encapsulate best-practice patterns and can be reused across projects with a simple interface.
3 	You are building CDK apps for dev, test, and prod accounts in different regions. What CDK feature helps you target the correct account and region per deployment?	A. Context B. Environment C. Bootstrap D. Synth	ANSWER: B Environments in CDK define the AWS account and region where the stack will be deployed.
4 	Your team wants to automatically deploy CDK stacks through a CI/CD pipeline with approval before promoting to production. What should you use?	A. AWS CodePipeline + CDK Pipelines B. AWS CodeBuild only C. AWS CodeDeploy D. AWS CloudFormation Change Sets	ANSWER: A CDK Pipelines (built on CodePipeline) support multi-stage workflows, approvals, and environment promotions.
5 	You need to pass values between two CDK stacks and import outputs from one stack into another. What is the BEST way to achieve this?	A. SSM Parameter Store B. CloudFormation Outputs + Fn.importValue C. Hardcode values D. Environment Variables	ANSWER: B CDK uses CloudFormation Outputs and Fn.importValue (or direct references within the same app) to share values between stacks.

BEST PRACTICES



- ✓ USE L3 CONSTRUCTS for common patterns.
- ✓ KEEP STACKS SMALL and focused on a single domain.
- ✓ USE CONTEXT for environment-specific configuration.
- ✓ BOOTSTRAP CDK in each target account/region.
- ✓ USE CDK PIPES for reliable multi-account deployments.
- ✓ REVIEW CDK DIFF before deploying changes.
- ✓ VERSION CONTROL all CDK code and lock dependencies.



PRO TIP

Always run "cdk synth" to see the generated CloudFormation template. If you wouldn't write it by hand, refine your constructs and abstractions!

CDK DEPLOYMENT WORKFLOW



WRITE CDK CODE
(TypeScript / Python)



SYNTHESIZE
(cdk synth)



DIFF CHANGES
(cdk diff)



DEPLOY
(cdk deploy)



MANAGE & MONITOR
(AWS Console / CW)



/ VERIQTA SOCIAL



VERIQTA



/ VERIQTA SOCIAL

AWS DEVOPS ENGINEER PROFESSIONAL CONFIGURATION MANAGEMENT

★ 5 SCENARIO QUESTIONS + ANSWERS ★

TOPIC OVERVIEW					
SYSTEMS MANAGER  - Central hub for operations - Manage EC2, on-prem, containers - Secure access & automation - Inventory & compliance	PARAMETER STORE  - Securely store configuration data - Standard & Secure String parameters - Versioning - IAM access control	STATE MANAGER  - Define desired state for instances - Automate configurations - Enforce compliance - Support for patching, policies	RUN COMMAND  - Run commands remotely - No inbound open ports - Output to S3 or CloudWatch Logs - Target by tags, instance, or group	AUTOMATION  - Automate multi-step tasks - Predefined or custom runbooks - Approval workflow support - Trigger by events or schedules	PATCH MANAGER  - Automate OS & application patching - Compliance reporting - Maintenance windows - Multi-account patching

5 SCENARIO QUESTIONS + ANSWERS

1 	You need a central place to manage EC2 instances, run automated tasks, and gain operational visibility without logging into instances. Which service should you use?	A. Parameter Store B. Systems Manager C. State Manager D. Run Command	ANSWER: B AWS Systems Manager provides a centralized console to manage, automate, and operate your infrastructure at scale.
2 	You need to store a database password securely and allow EC2 instances to retrieve it at runtime. Which service should you use?	A. S3 B. Parameter Store C. CloudTrail D. Secrets Manager	ANSWER: B Parameter Store securely stores configuration data and can be accessed by EC2 instances using IAM permissions.
3 	You must ensure that a specific security configuration is always applied to all EC2 instances and automatically remediated if changed. Which service is BEST?	A. State Manager B. Run Command C. Automation D. CloudWatch	ANSWER: A State Manager enforces the desired state by automatically applying configurations and remediating drift.
4 	You need to run a one-time script on multiple EC2 instances without opening SSH ports. Which service should you use?	A. Run Command B. State Manager C. Automation D. Patch Manager	ANSWER: A Run Command allows you to run commands remotely on multiple instances without requiring inbound access.
5 	You want to automatically install OS patches on EC2 instances during a maintenance window and generate compliance reports. Which service should you use?	A. Automation B. Patch Manager C. State Manager D. Parameter Store	ANSWER: B Patch Manager automates patch deployment, manages maintenance windows, and provides compliance reporting.

BEST PRACTICES



- ✓ Use **SYSTEMS MANAGER** as your operational hub.
- ✓ Store secrets in **PARAMETER STORE** or **SECRETS MANAGER**, not in code.
- ✓ Use **STATE MANAGER** to enforce configuration compliance.
- ✓ Use **RUN COMMAND** for ad-hoc or one-time tasks.
- ✓ Use **AUTOMATION** for repeatable, multi-step processes.
- ✓ Use **PATCH MANAGER** to keep systems updated and compliant.



PRO TIP

Combine these services: Use State Manager for compliance, Run Command for quick fixes, Automation for complex tasks, and Patch Manager for updates. This reduces manual work and improves reliability.

QUICK SERVICE GUIDE

Systems Manager	: Central management and automation
Parameter Store	: Secure configuration and secrets
State Manager	: Maintain desired state and compliance
Run Command	: Run commands on managed instances
Automation	: Automate complex, repeatable tasks
Patch Manager	: Automate patching and ensure compliance



REMEMBER

These services work best together. Build secure, automated, and self-healing infrastructure using AWS Systems Manager capabilities.

SYSTEMS MANAGER PRODUCTION OPERATIONS

5 SCENARIO QUESTIONS + ANSWERS



SESSION
MANAGER



FLEET
MANAGEMENT



AUTOMATION
DOCUMENTS



MAINTENANCE
WINDOWS



INVENTORY &
TROUBLESHOOTING



EXAM FOCUS

- Practical operations
- Troubleshooting
- Automation
- Reliability

1. A SECURITY ENGINEER NEEDS TO SECURELY ACCESS EC2 INSTANCES WITHOUT OPENING SSH PORTS. WHAT SHOULD THEY USE?

- A. SSH over bastion host
- B. Systems Manager Session Manager
- C. EC2 Instance Connect
- D. RDP over VPN



ANSWER: B

Session Manager provides secure, auditable SSH/CLI access without opening inbound ports or using bastion hosts.

WHY OTHERS ARE WRONG:

- A: Bastion hosts still expose SSH.
- C: Requires OS user management and is not fully auditable.
- D: RDP is for Windows and requires open ports or VPN.

2. A TEAM HAS 500 EC2 INSTANCES ACROSS MULTIPLE ACCOUNTS. THEY NEED A CENTRAL VIEW AND MANAGEMENT. WHAT SHOULD THEY USE?

- A. Systems Manager Explorer
- B. AWS Config
- C. CloudWatch Dashboard
- D. Trusted Advisor



ANSWER: A

Explorer provides aggregated operational data across accounts and Regions for managed instances.

WHY OTHERS ARE WRONG:

- B: Config tracks configuration, not operational view.
- C: CloudWatch shows metrics, not fleet operational data.
- D: Trusted Advisor provides recommendations only.

3. A DEVOPS ENGINEER WANTS TO RUN A REBOOT ACROSS ALL WINDOWS SERVERS DURING OFF-HOURS AUTOMATICALLY. WHAT IS THE BEST SOLUTION?

- A. Run Command manually
- B. Automation Document + Maintenance Window
- C. Lambda function
- D. EC2 Instance Connect



ANSWER: B

Use an Automation Document to reboot and schedule it using Maintenance Windows for controlled execution.

WHY OTHERS ARE WRONG:

- A: Manual execution does not scale.
- C: Lambda is not ideal for OS-level actions.
- D: Instance Connect is for access, not automation.

4. AN INSTANCE IS SLOW AND USERS COMPLAIN ABOUT HIGH CPU. HOW CAN YOU QUICKLY TROUBLESHOOT USING SYSTEMS MANAGER?

- A. Check CloudWatch metrics only
- B. Use Session Manager to connect and diagnose
- C. Rely on Trusted Advisor
- D. Stop and start the instance



ANSWER: B

Session Manager allows secure shell access to diagnose issues in real time without opening SSH ports.

WHY OTHERS ARE WRONG:

- A: Metrics show symptoms, not root cause.
- C: Trusted Advisor does not diagnose live issues.
- D: Stop/start can cause downtime and is not troubleshooting.

5. A COMPANY NEEDS TO TRACK INSTALLED SOFTWARE AND OS PATCH LEVELS ON ALL MANAGED INSTANCES. WHICH SYSTEMS MANAGER CAPABILITY SHOULD THEY USE?

- A. Inventory
- B. Run Command
- C. Parameter Store
- D. Automation Documents



ANSWER: A

Inventory collects configuration data including software inventory, patch levels, and system details.

WHY OTHERS ARE WRONG:

- B: Run Command executes commands but does not collect inventory by default.
- C: Parameter Store stores values, not inventory data.
- D: Automation Documents run tasks, not collect inventory.

KEY SYSTEMS MANAGER CAPABILITIES

CAPABILITY	USE CASE
SESSION MANAGER	Secure instance access without SSH ports
FLEET MANAGEMENT	Centralized view and management at scale
AUTOMATION DOCUMENTS	Automate operations and remediation
MAINTENANCE WINDOWS	Schedule changes during maintenance
INVENTORY	Track software, patches, and configurations



PRODUCTION TIP

Always combine Automation Documents with Maintenance Windows for predictable, safe, and repeatable operations.



EXAM STRATEGY

Focus on which service provides the MOST scalable, secure, and operationally efficient solution.

AMAZON CLOUDWATCH MONITORING

5 SCENARIO QUESTIONS + ANSWERS



METRICS



LOGS



ALARMS



DASHBOARDS



METRIC FILTERS



COMPOSITE ALARMS

EXAM FOCUS

- Monitor Everything
- Detect Issues Early
- Automate Alerts
- Operational Excellence

1. CPU UTILIZATION ON AN EC2 INSTANCE SPIKES ABOVE 85% FOR 10 MINUTES. YOU NEED AN ALARM TO NOTIFY THE TEAM. WHAT IS THE BEST APPROACH?

- A. Create a CloudWatch metric alarm on CPUUtilization
- B. Check logs in CloudWatch Logs Insights
- C. Create a metric filter and alarm on it
- D. Use AWS Trusted Advisor

✓ ANSWER: A

Create a CloudWatch metric alarm on CPUUtilization with a threshold of 85% for 10 minutes.

WHY OTHERS ARE WRONG:

- B: Logs track events, not real-time metrics.
- C: Metric filters are for logs, not native metrics.
- D: Trusted Advisor provides recommendations, not alerts.

2. YOUR APPLICATION LOGS CONTAIN ERROR MESSAGES. YOU NEED TO TRACK THE NUMBER OF ERROR MESSAGES OVER TIME. WHAT SHOULD YOU DO?

- A. Create a CloudWatch Logs metric filter for the error pattern
- B. Enable detailed monitoring for the instance
- C. Create a CloudWatch dashboard
- D. Create a composite alarm

✓ ANSWER: A

Create a metric filter on the log group to count error messages and publish the metric.

WHY OTHERS ARE WRONG:

- B: Detailed monitoring gives metrics, not log counts.
- C: Dashboards visualize; they don't extract metrics.
- D: Composite alarms combine alarms; they don't create metrics.

3. YOU NEED A SINGLE VIEW TO MONITOR CPU, MEMORY, DISK, AND REQUEST COUNT FOR YOUR APPLICATION. WHAT SHOULD YOU USE?

- A. CloudWatch Dashboard
- B. CloudWatch Alarm
- C. AWS Config
- D. CloudTrail

✓ ANSWER: A

CloudWatch Dashboards provide a centralized visual view of multiple metrics.

WHY OTHERS ARE WRONG:

- B: Alarms notify; they don't visualize.
- C: AWS Config tracks configuration, not metrics.
- D: CloudTrail records API activity.

4. YOU WANT TO TRIGGER AN ALARM ONLY IF CPU IS HIGH AND STATUS CHECK FAILURES ARE ALSO INCREASING. WHAT SHOULD YOU USE?

- A. Composite Alarm
- B. Metric Math Alarm
- C. Simple Alarm
- D. EventBridge Rule

✓ ANSWER: A

Composite Alarms combine multiple conditions and reduce alarm noise.

WHY OTHERS ARE WRONG:

- B: Metric Math is for calculations, not combining conditions.
- C: Simple alarms evaluate a single metric.
- D: EventBridge triggers on events, not metric conditions.

5. YOU DEPLOY A NEW VERSION AND NEED TO MONITOR 5XX ERRORS IN REAL TIME TO DETECT ISSUES EARLY. WHAT IS THE BEST SOLUTION?

- A. Create a metric filter for 5XX errors and set an alarm
- B. Check the application logs manually
- C. Use AWS Health Dashboard
- D. Rely on Auto Scaling

✓ ANSWER: A

Metric filter extracts 5XX errors from logs and the alarm notifies immediately.

WHY OTHERS ARE WRONG:

- B: Manual checks are slow and not scalable.
- C: Health Dashboard shows AWS service health.
- D: Auto Scaling reacts to load, not errors.

CLOUDWATCH MONITORING FLOW



KEY CLOUDWATCH COMPONENTS

	METRICS	Time-series data about performance and operational health.
	LOGS	Centralized log collection and analysis.
	ALARMS	Monitor metrics and perform actions when thresholds are breached.
	DASHBOARDS	Visualize metrics and logs in a single pane of glass.
	METRIC FILTERS	Extract structured data from logs and create metrics.
	COMPOSITE ALARMS	Combine multiple alarms to reduce noise and improve accuracy.



PRODUCTION TIP

Use metric filters + composite alarms to detect real issues, not just individual spikes. This reduces alert fatigue.



EXAM STRATEGY

Look for keywords: detect, alert, monitor, aggregate, combine conditions, visualize, count errors.

CLOUDWATCH LOGS AND OBSERVABILITY

5 SCENARIO QUESTIONS + ANSWERS



LOG GROUPS



LOG STREAMS



LOGS INSIGHTS



RETENTION

SUBSCRIPTION
FILTERSCENTRALIZED
LOGGING

1. YOUR APPLICATION IS RUNNING ON EC2 INSTANCES AND YOU NEED TO ORGANIZE LOGS BY ENVIRONMENT AND APPLICATION. HOW SHOULD YOU STRUCTURE CLOUDWATCH LOGS?

- A. Create one log group and one stream for everything
- B. Create separate log groups per environment and log streams per instance/application
- C. Create log groups per instance only
- D. Store logs directly in S3

✓ **ANSWER: B**

Use log groups to separate environments and log streams for each instance or application to improve organization.

WHY OTHERS ARE WRONG:

- A: Hard to manage and query.
- C: Does not separate environments.
- D: Loses CloudWatch Logs features.

3. YOUR LOGS ARE GROWING FAST AND YOU NEED TO CONTROL COSTS BY AUTOMATICALLY REMOVING OLD LOGS. WHAT SHOULD YOU DO?

- A. Export logs to S3 manually
- B. Set a retention policy on the log group
- C. Delete log streams manually
- D. Disable logging

✓ **ANSWER: B**

Set a retention policy on the log group to automatically delete logs after the defined number of days.

WHY OTHERS ARE WRONG:

- A: Manual process and still incurs storage cost.
- C: Not scalable and does not remove old logs.
- D: You lose all logs and visibility.

5. YOU NEED A CENTRALIZED LOGGING SOLUTION FOR MULTIPLE ACCOUNTS AND REGIONS. WHAT IS THE BEST APPROACH?

- A. Create separate log groups in each account
- B. Use CloudWatch cross-account subscriptions to a central account
- C. Use local S3 buckets in each account
- D. Disable logging in non-prod accounts

✓ **ANSWER: B**

Cross-account subscriptions send logs from source accounts and regions to a central logging account for analysis.

WHY OTHERS ARE WRONG:

- A: Hard to manage at scale.
- C: Not centralized and harder to query.
- D: Eliminates visibility and troubleshooting.

2. YOU NEED TO FIND ALL ERROR MESSAGES IN YOUR APPLICATION LOGS FROM THE LAST 2 HOURS. WHAT IS THE FASTEST WAY?

- A. Download logs and search manually
- B. Use CloudWatch Logs Insights
- C. Create a metric filter and alarm
- D. Use CloudTrail

✓ **ANSWER: B**

CloudWatch Logs Insights allows fast search and analysis of log data using a query language.

WHY OTHERS ARE WRONG:

- A: Slow and not scalable.
- C: Useful for metrics, not immediate search.
- D: CloudTrail logs API activity, not application logs.

4. YOU WANT TO SEND ALL APPLICATION LOGS TO AMAZON OPENSEARCH SERVICE IN REAL TIME FOR ADVANCED ANALYSIS. WHAT SHOULD YOU USE?

- A. Log export to S3
- B. Subscription filter
- C. CloudWatch agent
- D. Log Insights

✓ **ANSWER: B**

Subscription filters stream log events from CloudWatch Logs to destinations like OpenSearch in real time.

WHY OTHERS ARE WRONG:

- A: Delayed and not real-time.
- C: Collects logs but does not stream to OpenSearch.
- D: Queries logs but does not send them.

CLOUDWATCH LOGS FLOW



KEY CLOUDWATCH LOGS COMPONENTS

LOG GROUPS	Logical container for log streams. Set retention at group level.
LOG STREAMS	Ordered sequence of log events from a specific source.
LOGS INSIGHTS	Query logs using powerful SQL-like query language.
RETENTION	Automatically delete logs after defined number of days.
SUBSCRIPTION FILTERS	Stream logs in real time to AWS services or custom endpoints.
CENTRALIZED LOGGING	Aggregate logs across accounts and regions for visibility.



PRODUCTION TIP

Use structured logging (JSON) and Logs Insights queries for faster troubleshooting and better observability.



EXAM STRATEGY

Look for keywords: search, analyze, real-time, centralize, retention, stream, aggregate. Choose the MOST efficient solution.

EXAM FOCUS

- Centralized Logging
- Fast Troubleshooting
- Real-time Insights
- Observability at Scale

EVENT-DRIVEN OPERATIONS

5 SCENARIO QUESTIONS + ANSWERS



EVENTBRIDGE



EVENT RULES



EVENT PATTERNS



TARGETS



SCHEDULED EVENTS

AUTOMATED
REMEDiation

EXAM FOCUS

- Event-Driven Architecture
- Automation
- Reliable Remediation
- Scalable Integrations

1. YOU NEED TO TRIGGER A LAMBDA FUNCTION WHEN AN EC2 INSTANCE IS STOPPED. WHAT IS THE MOST EFFICIENT WAY?

- A. Create a CloudWatch alarm
- B. Use an EventBridge rule with an EC2 state-change event
- C. Use AWS Config rule
- D. Use a Systems Manager Automation

✓ ANSWER: B

EventBridge can match the EC2 state-change event and trigger the Lambda function immediately.

WHY OTHERS ARE WRONG:

- A: Alarms monitor metrics, not state changes.
- C: Config is for compliance, not real-time events.
- D: Automation is not triggered by EC2 events by default.

2. YOU NEED TO CAPTURE ALL API CALLS TO IAM AND SEND THEM TO AN SNS TOPIC FOR NOTIFICATIONS. HOW SHOULD YOU DO THIS?

- A. CloudWatch metric filter
- B. EventBridge rule with CloudTrail events
- C. S3 event notification
- D. AWS Config aggregation

✓ ANSWER: B

EventBridge can capture CloudTrail management events for IAM and send them to SNS.

WHY OTHERS ARE WRONG:

- A: Metric filters work on logs, not API events.
- C: S3 events are unrelated to IAM API calls.
- D: Config does not capture API call events.

3. YOU NEED TO RUN AN AUTOMATION DOCUMENT EVERY DAY AT 1:00 AM UTC TO CLEAN UP OLD SNAPSHOTS. WHICH EVENTBRIDGE FEATURE SHOULD YOU USE?

- A. Event pattern
- B. Schedule expression
- C. Dead-letter queue
- D. Archive and replay

✓ ANSWER: B

Schedule expressions in EventBridge rules allow you to run tasks on a fixed schedule.

WHY OTHERS ARE WRONG:

- A: Event patterns match events, not time.
- C: DLQ handles failed deliveries, not scheduling.
- D: Archive is for storing events, not triggering them.

4. YOU NEED TO SEND DIFFERENT EVENTS TO DIFFERENT TARGETS BASED ON THE EVENT TYPE. WHAT IS THE BEST APPROACH?

- A. Create multiple EventBridge rules with different event patterns
- B. Use one rule and filter in the target application
- C. Use CloudWatch Logs metric filters
- D. Use SNS filter policies

✓ ANSWER: A

EventBridge rules should be responsible for routing based on event patterns.

WHY OTHERS ARE WRONG:

- B: Puts unnecessary logic in the target.
- C: Metric filters are not for event routing.
- D: SNS filters work after delivery, not before.

5. YOU NEED TO AUTOMATICALLY RESTART AN EC2 INSTANCE IF CPU UTILIZATION IS ABOVE 90% FOR 5 MINUTES. WHAT IS THE BEST SOLUTION?

- A. CloudWatch alarm + EventBridge rule + SSM Automation
- B. EventBridge rule + Lambda + EC2 API
- C. Auto Scaling group
- D. Systems Manager Run Command manually

✓ ANSWER: A

Use a CloudWatch alarm to detect the issue, EventBridge to trigger, and SSM Automation to restart.

WHY OTHERS ARE WRONG:

- B: Lambda adds complexity and more failure points.
- C: Auto Scaling doesn't react to CPU on existing instance.
- D: Manual actions are not automated remediation.



PRODUCTION TIP

Keep event patterns specific to avoid noise.
Use automation for safe and repeatable remediation.

EVENTBRIDGE ARCHITECTURE FLOW



COMMON EVENTBRIDGE TARGETS

TARGET	USE CASE
LAMBDA	Run custom code in response to events
SNS	Send notifications to users or systems
SQS	Queue messages for decoupled processing
SYSTEMS MANAGER	Run Automation documents or Run Command
STEP FUNCTIONS	Start workflows for complex automation



EXAM STRATEGY

Focus on keywords: trigger, automate, schedule, route, pattern, remediate, notify. Choose the MOST scalable and automated solution.

AWS CLOUDTRAIL

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Audit and Compliance
- Security Monitoring
- Forensics and Investigation
- Event History Analysis



MANAGEMENT EVENTS



DATA EVENTS



INSIGHTS EVENTS



ORGANIZATION TRAILS



LOG INTEGRITY



API INVESTIGATION

1. YOU NEED TO KNOW WHO DELETED AN S3 BUCKET LAST NIGHT. WHICH CLOUDTRAIL EVENT TYPE WILL SHOW THIS ACTIVITY?

- A. Data event
- B. Management event
- C. Insights event
- D. Insights rule



ANSWER: B

Deleting an S3 bucket is a management event.

WHY OTHERS ARE WRONG:

- A: Data events capture object-level activity, not bucket deletion.
- C: Insights events detect unusual activity, not the event itself.
- D: Insights rules are detectors, not event types.

2. YOU NEED TO AUDIT WHO DOWNLOADED AN OBJECT FROM AN S3 BUCKET. WHAT SHOULD YOU ENABLE?

- A. Management events
- B. Data events for S3
- C. Insights events
- D. Organization trail only



ANSWER: B

S3 object-level operations (GetObject) are data events.

WHY OTHERS ARE WRONG:

- A: Does not include object-level API calls.
- C: Detects anomalies, not raw object access.
- D: Organization trails do not capture data events.

3. YOU WANT TO DETECT UNUSUAL API ACTIVITY, SUCH AS A SUDDEN SPIKE IN ERROR RATES. WHAT CLOUDTRAIL FEATURE SHOULD YOU USE?

- A. Management events
- B. Data events
- C. Insights events
- D. Log file validation



ANSWER: C

CloudTrail Insights automatically detects unusual activity patterns.

WHY OTHERS ARE WRONG:

- A: Records events but does not detect anomalies.
- B: Captures data events, not anomaly detection.
- D: Validates log integrity, not unusual activity.

5. YOU NEED TO ENSURE CLOUDTRAIL LOG FILES HAVE NOT BEEN TAMPERED WITH. WHAT FEATURE SHOULD YOU USE?

- A. CloudTrail Insights
- B. Log file validation (Digest files)
- C. Data events
- D. EventBridge rule



ANSWER: B

CloudTrail log file validation uses digest files to verify integrity.

WHY OTHERS ARE WRONG:

- A: Detects anomalies, not integrity.
- C: Captures events, not integrity.
- D: Routes events, not validation.

4. YOUR ORGANIZATION HAS MULTIPLE ACCOUNTS AND YOU NEED A CENTRAL TRAIL TO AUDIT ACTIVITY ACROSS ALL ACCOUNTS. WHAT SHOULD YOU DO?

- A. Create individual trails in each account
- B. Create an organization trail
- C. Enable CloudTrail Insights
- D. Use AWS Config



ANSWER: B

Organization trails capture events from all accounts in the organization.

WHY OTHERS ARE WRONG:

- A: Harder to manage and analyze.
- C: Detects anomalies, does not aggregate accounts.
- D: Tracks resource configuration, not API activity.

CLOUDTRAIL EVENT TYPES

	MANAGEMENT EVENTS	Record management operations performed on resources (e.g., RunInstances, DeleteBucket).
	DATA EVENTS	Record resource-level operations performed on or within a resource (e.g., GetObject, PutObject).
	INSIGHTS EVENTS	Detect unusual API activity patterns for faster threat detection.
	ORGANIZATION TRAILS	Capture events from all accounts in an AWS Organization.

CLOUDTRAIL KEY BENEFITS

	AUDIT API ACTIVITY		INVESTIGATE SECURITY INCIDENTS		MEET COMPLIANCE REQUIREMENTS		ENSURE LOG INTEGRITY		DETECT UNUSUAL BEHAVIOR
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PRODUCTION TIP

Enable organization trails and Insights events. Store logs in a dedicated S3 bucket with MFA delete and bucket versioning for maximum security.



EXAM STRATEGY

Pay attention to: event type, purpose, scope, and the MOST effective or secure solution.

AWS CONFIG

5 SCENARIO QUESTIONS + ANSWERS



EXAM FOCUS

- Continuous Compliance
- Automated Remediation
- Governance at Scale
- Visibility and Relationships



CONFIGURATION HISTORY



CONFIG RULES



CONFORMANCE PACKS



COMPLIANCE MONITORING



AUTOMATED REMEDIATION



RESOURCE RELATIONSHIPS

1. YOU NEED TO TRACK ALL CHANGES MADE TO AN EC2 INSTANCE OVER TIME. WHICH AWS CONFIG FEATURE SHOULD YOU USE?

- A. Config Rules
- B. Configuration History
- C. Conformance Packs
- D. Resource Relationships



ANSWER: B

Configuration History records all configuration changes for supported resources over time.

WHY OTHERS ARE WRONG:

- A: Rules evaluate compliance, not historical changes.
- C: Conformance Packs group rules, not history.
- D: Resource Relationships show connections, not changes.

2. YOU NEED TO ENSURE THAT ALL S3 BUCKETS HAVE VERSIONING ENABLED. WHICH CONFIG FEATURE SHOULD YOU USE?

- A. Configuration History
- B. Config Rule
- C. Resource Relationships
- D. Conformance Pack



ANSWER: B

A Config Rule can evaluate S3 buckets and mark them NON_COMPLIANT if versioning is not enabled.

WHY OTHERS ARE WRONG:

- A: History shows past changes, not compliance.
- C: Relationships do not evaluate configuration.
- D: Packs group rules, but rules do the evaluation.

3. YOU WANT TO DEPLOY AWS BEST PRACTICE RULES ACROSS ALL ACCOUNTS IN YOUR ORGANIZATION. WHAT IS THE MOST EFFICIENT WAY?

- A. Create individual rules in each account
- B. Use AWS Config Conformance Packs
- C. Use CloudTrail
- D. Use IAM Access Analyzer



ANSWER: B

Conformance Packs deploy a set of managed rules consistently across accounts and Regions.

WHY OTHERS ARE WRONG:

- A: Harder to manage and not scalable.
- C: CloudTrail records API activity, not compliance.
- D: Access Analyzer is for IAM permissions analysis.

4. YOUR TEAM NEEDS TO BE ALERTED WHEN A RESOURCE BECOMES NON_COMPLIANT. WHAT SHOULD YOU DO?

- A. Use Configuration History
- B. Create a Config Rule and set up an SNS notification
- C. Use Resource Relationships
- D. Enable AWS Trusted Advisor



ANSWER: B

Config Rules evaluate resources. You can trigger an SNS notification when a rule reports NON_COMPLIANT.

WHY OTHERS ARE WRONG:

- A: History does not trigger notifications.
- C: Relationships do not evaluate compliance.
- D: Trusted Advisor is for recommendations, not alerts.

AWS CONFIG PROCESS FLOW



KEY AWS CONFIG COMPONENTS

	CONFIGURATION HISTORY	Records and Stores changes to supported resources over time.
	CONFIG RULES	Evaluates resource configurations and determines compliance.
	CONFORMANCE PACKS	Groups related AWS managed rules for consistent compliance.
	COMPLIANCE MONITORING	Tracks compliance status and reporting across accounts and Regions.
	AUTOMATED REMEDIATION	Automatically fixes non-compliant resources using SSM Automation or Lambda.
	RESOURCE RELATIONSHIPS	Shows how resources are related and connected.



PRODUCTION TIP

Combine Config Rules, Conformance Packs, and Automated Remediation for continuous compliance and reduced manual effort.



EXAM STRATEGY

Focus on keywords: detect, monitor, enforce, automate, maintain, group, track, and relationship. Choose the MOST scalable and automated solution.

INCIDENT AND EVENT RESPONSE

5 SCENARIO QUESTIONS + ANSWERS



DETECTION



INVESTIGATION



CONTAINMENT

AUTOMATED
REMEDIATION

RECOVERY

POST-INCIDENT
ANALYSIS

EXAM FOCUS

- Fast Detection
- Effective Containment
- Automated Remediation
- Quick Recovery
- Post-Incident Improvement

1. YOUR APPLICATION IS EXPERIENCING HIGH LATENCY. CPU AND MEMORY ARE NORMAL. HOW DO YOU DETECT AND IDENTIFY THE ISSUE FASTEST?

- A. Amazon CloudWatch Dashboards
- B. CloudWatch Logs Insights
- C. AWS Config Rules
- D. AWS Trusted Advisor



ANSWER: B

CloudWatch Logs Insights can quickly search and identify latency-related errors or slow transactions.

WHY OTHERS ARE WRONG:

- A: Dashboards show metrics, not detailed root causes.
- C: Config checks compliance, not performance issues.
- D: Trusted Advisor gives recommendations, not live issues.

2. YOU SUSPECT A COMPROMISED IAM USER THAT CREATED AN ACCESS KEY. WHAT IS YOUR FIRST STEP IN INVESTIGATION?

- A. Disable the key immediately
- B. Check AWS CloudTrail for user activity
- C. Delete the user
- D. Rotate all IAM passwords



ANSWER: B

CloudTrail shows the full API activity and helps confirm what actions the user performed.

WHY OTHERS ARE WRONG:

- A: Act without evidence may disrupt legitimate users.
- C: Deleting may remove important evidence.
- D: Rotating all passwords is excessive and slow.

3. A MALICIOUS INSTANCE IS SPINNING UP NEW INSTANCES CONTINUOUSLY. WHAT SHOULD YOU DO FIRST TO CONTAIN THE ISSUE?

- A. Terminate the instance
- B. Isolate it using Security Groups or NACLs
- C. Reboot the instance
- D. Delete the Auto Scaling group



ANSWER: B

Isolating the instance stops further damage while preserving evidence for investigation.

WHY OTHERS ARE WRONG:

- A: Termination may stop evidence collection.
- C: Rebooting does not stop malicious behavior.
- D: Deleting ASG affects healthy instances too.

4. YOU WANT TO AUTOMATICALLY RESPOND TO HIGH CPU USAGE BY RESTARTING AN INSTANCE. WHAT IS THE BEST SOLUTION?

- A. Create a CloudWatch Alarm + EventBridge + SSM Automation
- B. Create a Lambda function and run it manually
- C. Create an AWS Config rule
- D. Use AWS Trusted Advisor



ANSWER: A

CloudWatch Alarm detects the issue, EventBridge triggers SSM Automation to restart the instance automatically.

WHY OTHERS ARE WRONG:

- B: Manual actions are not automated.
- C: Config rules do not perform remediation.
- D: Trusted Advisor does not take action.

5. AFTER RESOLVING AN INCIDENT, WHAT SHOULD YOU DO NEXT TO PREVENT FUTURE OCCURRENCES?

- A. Document the incident and conduct a post-incident review
- B. Delete old CloudWatch logs
- C. Ignore since the issue is resolved
- D. Reboot all servers



ANSWER: A

Post-incident analysis helps identify root causes, improve processes, and prevent recurrence.

WHY OTHERS ARE WRONG:

- B: Logs are important for future analysis.
- C: Skipping review leads to repeated incidents.
- D: Rebooting is not a preventive measure.



PRODUCTION TIP

Prepare runbooks, automate where possible, and test your incident response plan regularly.

INCIDENT RESPONSE FRAMEWORK



DETECTION

Detect issue using metrics, logs, alerts, or events.



INVESTIGATION

Analyze logs, metrics, and traces to find root cause.



CONTAINMENT

Isolate affected resources and stop further impact.

AUTOMATED
REMEDIATION

Use automation to fix or mitigate the issue fast.



RECOVERY

Restore normal operations and validate systems.

POST-INCIDENT
ANALYSIS

Review, learn, document, and improve.

KEY AWS SERVICES FOR INCIDENT RESPONSE

CLOUDWATCH
Metrics, Logs,
AlarmsCLOUDTRAIL
API Activity
HistoryEVENTBRIDGE
Event-driven
AutomationLAMBDA
Custom
AutomationSSM
AUTOMATION
Runbooks and
RemediationAWS CONFIG
Compliance
Monitoring

EXAM STRATEGY

Follow the incident lifecycle: detect, investigate, contain, remediate, recover, review. Choose the MOST effective and automated solution.

AUTOMATED REMEDIATION

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Automated Detection
- Automated Response
- Self-Healing Systems
- Reduce Manual Effort
- Improve Reliability



1. CPU UTILIZATION ON AN EC2 INSTANCE REMAINS ABOVE 90% FOR 5 MINUTES. HOW CAN YOU AUTOMATICALLY SCALE OUT TO PREVENT IMPACT?

- A. Create a Lambda function and run it manually
- B. Create a CloudWatch Alarm + EventBridge + Lambda to scale Auto Scaling group
- C. Use AWS Config rule to scale the Auto Scaling group
- D. Use Systems Manager Run Command

ANSWER: B

CloudWatch Alarm detects high CPU, EventBridge triggers Lambda to scale the Auto Scaling group automatically.

WHY OTHERS ARE WRONG:

- A: Manual action is not automated.
- C: Config rules evaluate, they do not scale resources.
- D: Run Command does not scale Auto Scaling groups.

2. AN S3 BUCKET IS CREATED WITHOUT VERSIONING ENABLED. HOW CAN YOU AUTOMATICALLY ENABLE VERSIONING?

- A. Create a CloudWatch Alarm
- B. Create an AWS Config rule + Auto Remediation
- C. Use Lambda function manually
- D. Use IAM Access Analyzer

ANSWER: B

AWS Config rule detects non-compliant bucket and auto remediation enables versioning.

WHY OTHERS ARE WRONG:

- A: Alarms do not detect configuration compliance.
- C: Manual Lambda is not automatic.
- D: Access Analyzer does not fix configuration issues.

3. A TARGET GROUP HAS UNHEALTHY TARGETS CONTINUOUSLY. HOW CAN YOU AUTOMATICALLY REPLACE UNHEALTHY INSTANCES?

- A. CloudWatch Alarm + EventBridge + Systems Manager Automation
- B. Disable the target group
- C. Create a Config rule
- D. Increase the health check timeout

ANSWER: A

Alarm detects unhealthy targets, EventBridge triggers SSM Automation to terminate bad instances and allow ASG to replace.

WHY OTHERS ARE WRONG:

- B: Does not replace instances, only stops traffic.
- C: Config evaluates, it does not terminate instances.
- D: Does not remove unhealthy instances.

4. YOU NEED A SELF-HEALING SOLUTION THAT RESTARTS AN EC2 INSTANCE IF A CRITICAL SERVICE STOPS. WHAT IS THE BEST APPROACH?

- A. CloudWatch Alarm + EventBridge + Systems Manager Run Command
- B. Config rule remediation
- C. Manual SSH restart
- D. Lambda function with cron job

ANSWER: A

Alarm detects the issue, EventBridge triggers SSM Run Command to restart the service or instance automatically.

WHY OTHERS ARE WRONG:

- B: Config remediation is for configuration drift, not service health.
- C: Manual process is not automated.
- D: Cron job is not event-driven and not reliable.

5. A DATABASE CPU SPIKES AT NIGHT DUE TO A SCHEDULED JOB. HOW CAN YOU AUTOMATICALLY THROTTLE THE JOB AND NOTIFY THE TEAM?

- A. CloudWatch Alarm + EventBridge + Lambda + SNS
- B. Config rule only
- C. IAM policy change
- D. Manually stop the job

ANSWER: A

Alarm detects CPU spike, EventBridge triggers Lambda to throttle the job and send SNS notification.

WHY OTHERS ARE WRONG:

- B: Config does not monitor performance metrics.
- C: IAM policies are unrelated to job throttling.
- D: Manual action is not automated.

AUTOMATED REMEDIATION ARCHITECTURE



COMMON REMEDIATION ACTIONS



PRODUCTION TIP

Automate everything you can, but always test remediation workflows in a safe environment first.



EXAM STRATEGY

Look for keywords: automatic, detect, trigger, remediate, notify, restart, scale, enforce, fix, self-healing. Choose the MOST automated and reliable solution.

IAM FOR DEVOPS

5 SCENARIO QUESTIONS + ANSWERS



IAM ROLES



POLICIES

PERMISSION
BOUNDARIES

STS

TEMPORARY
CREDENTIALSLEAST
PRIVILEGE

EXAM FOCUS

- Secure Access
- Temporary Credentials
- Least Privilege
- Permission Boundaries
- Identity for Workloads

1. YOUR CI/CD PIPELINE NEEDS TO DEPLOY APPLICATIONS TO AWS WITHOUT USING LONG-TERM ACCESS KEYS. WHAT IS THE BEST SOLUTION?

- A. Create a new IAM user with programmatic access
- B. Use an IAM Role assumed by the pipeline
- C. Store access keys in AWS Secrets Manager
- D. Use an IAM Group with AdministratorAccess



ANSWER: B

IAM Roles allow the pipeline to assume the role and get temporary credentials securely.

WHY OTHERS ARE WRONG:

- A: IAM users with keys are not recommended for CI/CD.
- C: Secrets Manager stores secrets, not a best practice here.
- D: Groups do not provide authentication or temporary access.

2. A DEVELOPER CAN PERFORM MORE ACTIONS THAN INTENDED WITH AN ATTACHED POLICY. WHAT IAM FEATURE CAN YOU USE TO LIMIT THE MAXIMUM PERMISSIONS?

- A. Inline Policy
- B. Permission Boundary
- C. SCP (Service Control Policy)
- D. Session Policy



ANSWER: B

Permission Boundaries set the maximum permissions an identity-based policy can allow.

WHY OTHERS ARE WRONG:

- A: Inline policies grant permissions; they don't restrict.
- C: SCPs apply at the organization level, not per identity.
- D: Session policies are temporary, not a hard boundary.

3. YOU NEED TO ALLOW CROSS-ACCOUNT ACCESS SO A TEAM IN ACCOUNT B CAN DEPLOY TO RESOURCES IN ACCOUNT A. WHAT SHOULD YOU DO?

- A. Create an IAM user in Account A and share the keys
- B. Create an IAM Role in Account A and trust Account B
- C. Add Account B to a group in Account A
- D. Enable access keys for the root user



ANSWER: B

Cross-account access should be done using IAM Roles with a trust relationship.

WHY OTHERS ARE WRONG:

- A: Sharing users or keys is insecure and not recommended.
- C: Groups do not provide cross-account access.
- D: Root user keys should never be used.

4. YOUR APPLICATION RUNNING ON EC2 NEEDS ACCESS TO S3 AND DYNAMODB. WHAT IS THE BEST WAY TO PROVIDE THESE PERMISSIONS?

- A. Attach a policy directly to the EC2 instance
- B. Store access keys in the application configuration
- C. Attach an IAM Role to the EC2 instance profile
- D. Create an IAM user for the application



ANSWER: C

Instance Profiles with IAM Roles provide temporary credentials to EC2 securely.

WHY OTHERS ARE WRONG:

- A: Policies cannot be attached directly to EC2 instances.
- B: Hardcoded keys are insecure and must be avoided.
- D: Users are for humans, not applications.

5. A LAMBDA FUNCTION NEEDS TO ACCESS AWS SERVICES. WHAT IS THE RECOMMENDED IAM APPROACH?

- A. Create an IAM user and embed keys in the code
- B. Attach an IAM Role to the Lambda function
- C. Use the AdministratorAccess managed policy
- D. Share the root user credentials



ANSWER: B

Lambda execution roles provide temporary credentials and follow least privilege.

WHY OTHERS ARE WRONG:

- A: Embedding keys is insecure and not recommended.
- C: AdministratorAccess violates least privilege.
- D: Root credentials must never be used.

IAM CONCEPT QUICK REFERENCE

	IAM ROLES	Identities that AWS services or users can assume to get temporary credentials.
	POLICIES	JSON documents that define permissions.
	PERMISSION BOUNDARIES	Set the maximum permissions an identity can have.
	STS (SECURITY TOKEN SERVICE)	Allows users or roles to get temporary security credentials.
	TEMPORARY CREDENTIALS	Short-lived credentials that reduce long-term security risks.
	LEAST PRIVILEGE	Grant only the minimum permissions required to perform a task.

AUTHENTICATION vs AUTHORIZATION

AUTHENTICATION (WHO YOU ARE)



Proves identity using credentials (users, roles, MFA, tokens).

AUTHORIZATION (WHAT YOU CAN DO)



Determines what actions you can perform on which resources (based on policies).



PRODUCTION TIP

Always follow least privilege, use roles for workloads, and never store long-term credentials in code or CI/CD.



EXAM STRATEGY

Focus on: roles vs users, permission boundaries, STS, temporary credentials, and least privilege principles. Eliminate unsafe or non-secure options.

CROSS-ACCOUNT ACCESS

5 SCENARIO QUESTIONS + ANSWERS



ASSUME ROLE



TRUST POLICIES



RESOURCE POLICIES



CROSS-ACCOUNT PIPELINES



MULTI-ACCOUNT DEPLOYMENTS



ORGANIZATIONS

EXAM FOCUS

- Secure Access
- Cross-Account Security
- Scalable Architectures
- Centralized Governance
- Least Privilege

1. A USER IN ACCOUNT A NEEDS TO DEPLOY RESOURCES IN ACCOUNT B USING TEMPORARY CREDENTIALS. WHAT IS THE BEST APPROACH?

- A. Create a new IAM user in Account B
- B. Share access keys between accounts
- C. Use AWS STS AssumeRole
- D. Enable root access in Account B

✓ ANSWER: C

AssumeRole from STS allows a user in Account A to temporarily assume a role in Account B securely.

WHY OTHERS ARE WRONG:

- A: Not scalable and hard to manage.
- B: Sharing keys is insecure and not recommended.
- D: Root access should never be enabled or used.

2. YOU NEED TO ALLOW ACCOUNT A TO ASSUME A ROLE IN ACCOUNT B. WHICH POLICY DO YOU MODIFY IN ACCOUNT B?

- A. IAM Identity policy
- B. Role trust policy
- C. Resource-based policy on S3
- D. Permission boundary

✓ ANSWER: B

The role trust policy in Account B defines which external accounts can assume the role.

WHY OTHERS ARE WRONG:

- A: Identity policies don't grant access to external accounts.
- C: Resource policies control resource access, not role assumption.
- D: Permission boundaries limit, not grant, access.

3. YOU NEED TO GRANT ACCOUNT B READ ACCESS TO AN S3 BUCKET IN ACCOUNT A. WHAT IS THE BEST SOLUTION?

- A. Create an IAM user in Account B
- B. Attach a bucket policy in Account A
- C. Share access keys for an IAM user
- D. Create a VPC peering connection

✓ ANSWER: B

A resource policy on the S3 bucket in Account A can grant Account B access directly.

WHY OTHERS ARE WRONG:

- A: Not required and not scalable.
- C: Sharing keys is insecure and not recommended.
- D: VPC peering is for network connectivity, not permissions.

4. YOUR CI/CD PIPELINE IN ACCOUNT A NEEDS TO DEPLOY TO AN ECS CLUSTER IN ACCOUNT B. WHAT IS THE BEST APPROACH?

- A. Store Account B credentials in plaintext in the pipeline
- B. Use cross-account role assumption in the pipeline
- C. Create a long-term IAM user in Account B
- D. Use the root account in Account B

✓ ANSWER: B

The pipeline should assume a role in Account B using STS for secure, temporary access.

WHY OTHERS ARE WRONG:

- A: Storing credentials is insecure.
- C: Long-term users are not ideal for automation.
- D: Root account must never be used.

5. YOU NEED TO MANAGE MULTIPLE AWS ACCOUNTS AT SCALE WITH CENTRALIZED GOVERNANCE. WHAT IS THE BEST SERVICE?

- A. AWS IAM
- B. AWS Organizations
- C. AWS Config
- D. AWS CloudTrail

✓ ANSWER: B

AWS Organizations enables centralized management, policy enforcement, and consolidated billing.

WHY OTHERS ARE WRONG:

- A: IAM manages permissions within a single account.
- C: Config monitors resources, not manage accounts.
- D: CloudTrail logs API activity, not governance.

CROSS-ACCOUNT ACCESS FLOW



KEY PRINCIPLES



PRODUCTION TIP

Always use roles with short-term credentials. Never share access keys. Audit and monitor all cross-account access.



EXAM STRATEGY

Focus on: AssumeRole, trust policies, resource policies, pipelines, Organizations, and least privilege. Choose the MOST secure, scalable, and maintainable solution.

SECRETS AND ENCRYPTION PRODUCTION OPERATIONS

5 SCENARIO QUESTIONS + ANSWERS



SECRETS
MANAGER



PARAMETER
STORE



AWS
KMS



KEY
POLICIES



ROTATION



CI/CD SECRET
PROTECTION



EXAM FOCUS

- Secure Secrets
- Encryption Best Practices
- Key Management
- Rotation & Automation
- CI/CD Protection

1. YOU NEED TO STORE DATABASE CREDENTIALS THAT APPLICATIONS RETRIEVE AT RUNTIME AND AUTOMATICALLY ROTATE. WHAT IS THE BEST SOLUTION?

- A. Store in EC2 user data
- B. Store in AWS Secrets Manager
- C. Store in S3 with encryption
- D. Store in Systems Manager Inventory



ANSWER: B

Secrets Manager is designed for sensitive data with built-in rotation and fine-grained access control.

WHY OTHERS ARE WRONG:

- A: User data is visible and not secure.
- C: S3 is not designed for secret retrieval at runtime.
- D: Inventory is for resource data, not secrets.

2. YOU NEED TO STORE A CONFIGURATION VALUE THAT IS NOT SENSITIVE AND WILL NEVER CHANGE. WHAT IS THE MOST COST-EFFECTIVE SOLUTION?

- A. AWS Secrets Manager
- B. SSM Parameter Store Standard Tier
- C. SSM Parameter Store Advanced Tier
- D. Store in a .env file in the application



ANSWER: B

Parameter Store Standard Tier is best for non-sensitive values and is free.

WHY OTHERS ARE WRONG:

- A: Secrets Manager costs more and is for secrets.
- C: Advanced Tier is for sensitive/encrypted values.
- D: Not secure and hard to manage.

3. YOUR APPLICATION NEEDS TO ENCRYPT SENSITIVE DATA BEFORE STORING IT IN S3. WHAT IS THE BEST APPROACH?

- A. Use SSE-S3 (AWS managed keys)
- B. Use AWS KMS customer managed key (CMK)
- C. Use client-side encryption with a hardcoded key
- D. Disable encryption for performance



ANSWER: B

KMS CMK gives full control, auditability, rotation, and fine-grained access using key policies.

WHY OTHERS ARE WRONG:

- A: SSE-S3 is less control and less audit visibility.
- C: Hardcoded keys are insecure and non-compliant.
- D: Never disable encryption for performance.

4. A CI/CD PIPELINE NEEDS ACCESS TO SECRETS DURING BUILD AND DEPLOYMENT. WHAT IS THE MOST SECURE SOLUTION?

- A. Store secrets in pipeline variables (plaintext)
- B. Store in Secrets Manager and grant access via IAM role
- C. Store in Parameter Store and hardcode IAM keys
- D. Commit secrets in encrypted files in the repo



ANSWER: B

Use Secrets Manager with least-privilege IAM role for the pipeline. No static credentials.

WHY OTHERS ARE WRONG:

- A: Plaintext variables are visible in logs and UI.
- C: Hardcoding credentials is insecure.
- D: Secrets in repo create high risk and compliance issues.

5. USERS ARE UNABLE TO DECRYPT DATA WITH A KMS KEY EVEN THOUGH THEY HAVE THE IAM PERMISSION `kms:Decrypt`. WHAT SHOULD YOU CHECK?

- A. The bucket policy
- B. The key policy
- C. The security group
- D. The subnet ACL



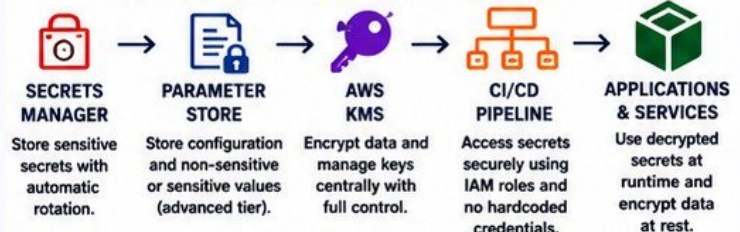
ANSWER: B

KMS permission is granted by the key policy, not just IAM policies.

WHY OTHERS ARE WRONG:

- A: Bucket policy does not control KMS usage.
- C: Security group does not affect KMS.
- D: Network ACLs do not control KMS permissions.

HOW THESE SERVICES WORK TOGETHER



KEY POINTS

- ✓ Use Secrets Manager for secrets that need rotation or dynamic credentials.
- ✓ Use Parameter Store for configuration and non-sensitive values.
- ✓ Use KMS CMK for encryption control, audits, and fine-grained access.
- ✓ Key policies control access. IAM alone is not enough.
- ✓ Never store secrets in code, environment variables, or public places.
- ✓ Use IAM roles in CI/CD. Avoid long-term access keys.

COMMON MISTAKES

- ✗ Using Secrets Manager for simple configuration.
- ✗ Storing secrets in environment variables.
- ✗ Hardcoding KMS keys or access keys.
- ✗ Not enabling rotation for database credentials.
- ✗ Using default KMS keys for sensitive data.
- ✗ Overly permissive key policies (* access).



PRODUCTION TIP

Enable rotation, use least privilege, monitor access in CloudTrail, and audit key usage regularly.



EXAM STRATEGY

Look for keywords: secure, least privilege, rotation, encryption, audit, cost-effective, and runtime access.

AMAZON S3 FOR DEVOPS WORKLOADS

5 SCENARIO QUESTIONS + ANSWERS



VERSIONING

LIFECYCLE
POLICIES

REPLICATION



ENCRYPTION

ARTIFACT
STORAGEACCESS
CONTROLS

EXAM FOCUS

- Secure & Scalable Storage
- Cost Optimization
- Data Protection
- DevOps Best Practices
- High Availability

1. YOU NEED TO PROTECT OBJECTS FROM ACCIDENTAL DELETION AND OVERWRITES. WHAT SHOULD YOU ENABLE?

- A. S3 Lifecycle Policy
- B. S3 Versioning
- C. S3 Replication
- D. S3 Object Lock



ANSWER: B

Versioning keeps multiple versions of an object so you can recover from accidental deletes or overwrites.

WHY OTHERS ARE WRONG:

- A: Lifecycle moves/archives objects, does not protect.
- C: Replication copies data, does not keep versions.
- D: Object Lock prevents deletion, not overwrites.

2. YOU WANT TO AUTOMATICALLY MOVE OLD LOG FILES TO GLACIER AFTER 30 DAYS AND DELETE THEM AFTER 365 DAYS. WHAT SHOULD YOU USE?

- A. S3 Replication
- B. S3 Lifecycle Policy
- C. S3 Versioning
- D. S3 Inventory



ANSWER: B

Lifecycle Policy can transition objects to Glacier and expire them after a defined period.

WHY OTHERS ARE WRONG:

- A: Replication copies, does not expire or transition.
- C: Versioning keeps multiple versions.
- D: Inventory generates reports, not lifecycle actions.

3. YOUR COMPANY NEEDS TO KEEP A COPY OF ALL S3 DATA IN ANOTHER AWS REGION FOR DR. WHAT SHOULD YOU CONFIGURE?

- A. S3 Lifecycle Policy
- B. S3 Cross-Region Replication (CRR)
- C. S3 Inventory
- D. S3 Access Points



ANSWER: B

CRR automatically replicates objects to a bucket in another AWS Region.

WHY OTHERS ARE WRONG:

- A: Lifecycle manages storage classes/expiration.
- C: Inventory provides reports, not data copies.
- D: Access Points manage access, not replication.

5. YOU NEED TO STORE BUILD ARTIFACTS FOR A CI/CD PIPELINE AND RESTRICT ACCESS SO ONLY THE PIPELINE ROLE CAN READ/WRITE. WHAT IS THE BEST APPROACH?

- A. Make the bucket public
- B. Use Bucket Policy with least-privilege IAM role
- C. Use pre-signed URLs for everyone
- D. Disable all access to the bucket



ANSWER: B

Bucket Policy + IAM role ensures least-privilege access for the pipeline only.

WHY OTHERS ARE WRONG:

- A: Public access is insecure.
- C: Pre-signed URLs are temporary, not ideal for pipelines.
- D: Would block the pipeline from accessing artifacts.

4. YOU NEED TO ENSURE ALL OBJECTS IN THE BUCKET ARE ENCRYPTED AT REST USING AWS KMS KEYS YOU MANAGE. WHAT SHOULD YOU DO?

- A. Use SSE-S3
- B. Use SSE-KMS
- C. Use Client-Side Encryption Only
- D. Disable Encryption



ANSWER: B

SSE-KMS uses AWS KMS keys you manage for encryption at rest with full control and audit.

WHY OTHERS ARE WRONG:

- A: SSE-S3 uses AWS managed keys, not customer keys.
- C: Client-side encryption is complex and key management is outside AWS.
- D: Not secure; data is stored unencrypted.

TYPICAL DEVOPS S3 WORKFLOW



S3 FEATURES QUICK REFERENCE

FEATURE	USE CASE
VERSIONING	Protect against accidental deletes/overwrites
LIFECYCLE POLICIES	Transition objects & expire old data
REPLICATION	Cross-Region or Cross-Account data copies
ENCRYPTION	Protect data at rest using SSE-S3 or SSE-KMS
ARTIFACT STORAGE	Store build artifacts, packages, containers, logs
ACCESS CONTROLS	IAM policies, bucket policies, Access Points



PRODUCTION TIP

Enable versioning, use lifecycle policies, enforce encryption, and restrict access by default. Automate S3 configurations with IaC for consistency.



EXAM STRATEGY

Look for keywords: protect, replicate, archive, encrypt, access control, cost, version, lifecycle, artifact.

EC2 DEPLOYMENT AND OPERATIONS

5 SCENARIO QUESTIONS + ANSWERS



AUTO
SCALING



LAUNCH
TEMPLATES



AMIS



INSTANCE
REPLACEMENT



HEALTH
CHECKS



DEPLOYMENT
FAILURES



EXAM FOCUS

- Reliable Deployments
- Operational Excellence
- Failure Recovery
- High Availability
- Cost Efficiency

1. YOUR APPLICATION IS EXPERIENCING HIGH TRAFFIC AND CPU UTILIZATION IS CONSISTENTLY ABOVE 80%. YOU NEED EC2 INSTANCES TO SCALE AUTOMATICALLY. WHAT SHOULD YOU USE?

- A. Add more instances manually
- B. Enable Auto Scaling with a target tracking policy
- C. Increase the instance size
- D. Use Elastic Load Balancer only



ANSWER: B

Auto Scaling with a target tracking policy automatically adjusts capacity based on CPU utilization.

WHY OTHERS ARE WRONG:

- A: Manual scaling is not automatic and not scalable.
- C: Bigger instances do not handle traffic spikes.
- D: ELB distributes traffic but does not add capacity.

2. YOU NEED TO DEPLOY EC2 INSTANCES WITH A STANDARDIZED CONFIGURATION THAT CAN BE UPDATED WITHOUT CREATING A NEW AMI EACH TIME. WHAT SHOULD YOU USE?

- A. AMI
- B. Launch Template
- C. User Data script
- D. EC2 Instance Connect



ANSWER: B

Launch Templates let you manage instance configuration independently of the Auto Scaling Group.

WHY OTHERS ARE WRONG:

- A: AMIs are static and require rebuilding for changes.
- C: User Data runs only at instance launch.
- D: EC2 Instance Connect is for access, not configuration.

3. YOUR APPLICATION SERVERS NEED TO BE REPLACED WITH NEW INSTANCES USING THE LATEST AMI WITH MINIMAL DOWNTIME. WHAT IS THE BEST APPROACH?

- A. Stop all instances and replace
- B. Terminate and recreate manually
- C. Use Instance Refresh in Auto Scaling
- D. Create a new Auto Scaling Group



ANSWER: C

Instance Refresh replaces instances one at a time with new AMIs while maintaining capacity.

WHY OTHERS ARE WRONG:

- A: Causes downtime.
- B: Manual process, risky and error-prone.
- D: Creates duplicate infrastructure.

5. A DEPLOYMENT FAILED DUE TO AN APPLICATION ERROR ON NEW INSTANCES. YOU NEED TO ROLL BACK TO THE PREVIOUS STABLE VERSION AUTOMATICALLY. WHAT SHOULD YOU IMPLEMENT?

- A. Manual rollback
- B. Use AMI rollback with Auto Scaling Lifecycle Hooks
- C. Disable health checks
- D. Increase termination policies



ANSWER: B

Lifecycle Hooks allow validation before instance termination and enable rollback to the previous stable version.

WHY OTHERS ARE WRONG:

- A: Slow and not automatic.
- C: Disabling health checks hides problems.
- D: Termination policies do not support rollback.

4. INSTANCES IN YOUR AUTO SCALING GROUP ARE MARKED UNHEALTHY AND GET TERMINATED REPEATEDLY. WHAT SHOULD YOU CHECK FIRST?

- A. Increase instance size
- B. Check health check configuration
- C. Disable Auto Scaling
- D. Change the AMI



ANSWER: B

Health check settings determine if instances are healthy. Incorrect checks cause replacements.

WHY OTHERS ARE WRONG:

- A: Does not address health check failures.
- C: Disabling does not fix the root cause.
- D: Not necessary unless there is an AMI issue.

EC2 DEPLOYMENT WORKFLOW



KEY EC2 CONCEPTS QUICK REFERENCE

CONCEPT	BEST PRACTICE
AUTO SCALING	Use target tracking for dynamic scaling.
LAUNCH TEMPLATES	Manage instance configuration and versions.
AMIS	Use latest stable AMI for deployments.
INSTANCE REFRESH	Update instances with minimal interruption.
HEALTH CHECKS	Configure ELB + EC2 health checks properly.
FAILURE HANDLING	Use lifecycle hooks and monitoring for rollback.



PRODUCTION TIP

Always monitor CloudWatch metrics, set proper health checks, and test deployments in staging before production.



EXAM STRATEGY

Look for keywords: scale, deploy, replace, healthy, rollback, minimal downtime, automatic, configuration.

AUTO SCALING AND RESILIENCE

5 SCENARIO QUESTIONS + ANSWERS



SCALING
POLICIES



TARGET
TRACKING



STEP
SCALING



LIFECYCLE
HOOKS



INSTANCE
REFRESH



FAILURE
RECOVERY



EXAM FOCUS

- Scale Automatically
- Stay Available
- Recover Fast
- Optimize Cost
- Handle Failures

1. YOUR APPLICATION TRAFFIC IS GRADUALLY INCREASING THROUGHOUT THE DAY. YOU WANT TO MAINTAIN 60% AVERAGE CPU UTILIZATION. WHAT IS THE BEST APPROACH?

- A. Step Scaling
- B. Target Tracking Scaling Policy
- C. Predictive Scaling
- D. Manual Scaling



ANSWER: B

Target Tracking maintains the average CPU around the desired value (60%) automatically.

WHY OTHERS ARE WRONG:

- A: Step Scaling reacts in fixed steps, not continuously.
- C: Predictive needs historical data and is not instant.
- D: Manual scaling is not automatic and not efficient.

2. YOUR TRAFFIC SPIKES SUDDENLY FROM 100 TO 1,000 REQUESTS PER MINUTE. YOU NEED THE FASTEST WAY TO SCALE OUT CAPACITY. WHAT SHOULD YOU USE?

- A. Target Tracking
- B. Step Scaling
- C. Instance Refresh
- D. Scheduled Scaling



ANSWER: B

Step Scaling adds a large number of instances quickly based on defined thresholds.

WHY OTHERS ARE WRONG:

- A: Target Tracking may scale more slowly.
- C: Instance Refresh updates instances, not scale out.
- D: Scheduled Scaling is time-based, not traffic-based.

3. YOU NEED TO PERFORM A DATABASE MIGRATION ON EC2 INSTANCES BEFORE TERMINATING THEM. WHAT SHOULD YOU CONFIGURE?

- A. Termination Policy
- B. Lifecycle Hook on Terminating
- C. Health Check Grace Period
- D. Cooldown Period



ANSWER: B

Lifecycle Hooks allow custom actions before an instance is terminated or put into service.

WHY OTHERS ARE WRONG:

- A: Termination Policy decides which instance to terminate.
- C: Grace period delays health checks, not termination.
- D: Cooldown period prevents scaling actions temporarily.

4. YOU WANT TO ROLL OUT A NEW AMI TO ALL INSTANCES WITH MINIMUM DOWNTIME AND AUTOMATIC ROLLBACK IF ISSUES OCCUR. WHAT SHOULD YOU USE?

- A. Replace all instances manually
- B. Instance Refresh with rollback enabled
- C. Detach and attach instances
- D. Stop and start instances



ANSWER: B

Instance Refresh replaces instances using the new AMI gradually and supports automatic rollback.

WHY OTHERS ARE WRONG:

- A: Manual replacement is risky and time-consuming.
- C: Detach/attach causes downtime.
- D: Stop/start does not update the AMI.

5. YOUR AUTO SCALING GROUP INSTANCES ARE FREQUENTLY MARKED UNHEALTHY AND TERMINATED DUE TO APPLICATION CRASHES. WHAT IS THE BEST APPROACH TO IMPROVE RESILIENCE?

- A. Increase Desired Capacity
- B. Check and fix the root cause + adjust health checks
- C. Disable Health Checks
- D. Use larger instance types



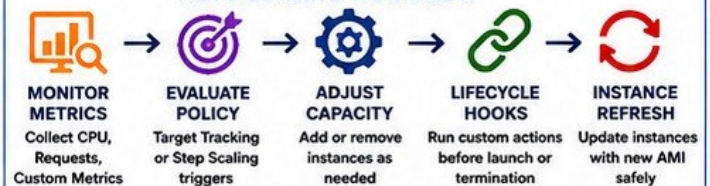
ANSWER: B

Fix the underlying issue and ensure health checks are configured correctly to prevent unnecessary terminations.

WHY OTHERS ARE WRONG:

- A: More bad instances do not fix the problem.
- C: Disabling health checks hides the problem.
- D: Bigger instances do not solve application failures.

AUTO SCALING WORKFLOW



SCALING POLICIES QUICK REFERENCE

POLICY	BEST USE CASE
TARGET TRACKING	Maintain a specific metric at a desired value (e.g., CPU 60%).
STEP SCALING	Scale based on metric thresholds in fixed steps.
SCHEDULED SCALING	Scale at specific times or recurring schedules.
PREDICTIVE SCALING	Use machine learning to predict and scale ahead.



PRODUCTION TIP

Combine the right scaling policy with proper health checks, lifecycle hooks, and instance refresh for resilient and cost-effective deployments.



EXAM STRATEGY

Look for keywords: spike, maintain, fastest, rollout, rollback, terminate, health check, update, replace, resilient.

ELASTIC LOAD BALANCING

5 SCENARIO QUESTIONS + ANSWERS



ALB



NLB



TARGET GROUPS



HEALTH CHECKS



LISTENER RULES



TRAFFIC SHIFTING



EXAM FOCUS

- Distribute Traffic Smartly
- Improve Availability
- Health & Performance
- Zero-Downtime Deployments
- Cost-Effective Architectures

1. YOUR WEB APPLICATION NEEDS TO ROUTE TRAFFIC BASED ON URL PATHS. WHICH LOAD BALANCER SHOULD YOU USE?

- A. Network Load Balancer (NLB)
- B. Application Load Balancer (ALB)
- C. Gateway Load Balancer (GWLB)
- D. Classic Load Balancer (CLB)



ANSWER: B

ALB operates at Layer 7 and supports path-based routing.

WHY OTHERS ARE WRONG:

- A: NLB works at Layer 4 and doesn't understand HTTP paths.
- C: GWLB is for third-party virtual appliances.
- D: CLB is legacy and doesn't support advanced routing.

2. YOUR APPLICATION NEEDS ULTRA-LOW LATENCY WITH MILLIONS OF REQUESTS PER SECOND. WHICH LOAD BALANCER IS THE BEST CHOICE?

- A. Application Load Balancer (ALB)
- B. Network Load Balancer (NLB)
- C. Classic Load Balancer (CLB)
- D. Gateway Load Balancer (GWLB)



ANSWER: B

NLB operates at Layer 4, is highly performant, and scales to millions of requests per second.

WHY OTHERS ARE WRONG:

- A: ALB is Layer 7 and adds more latency.
- C: CLB is legacy and not optimized for high performance.
- D: GWLB is for inspecting traffic, not for performance.

3. YOUR TARGET INSTANCES ARE UNHEALTHY AND TRAFFIC IS STILL BEING SENT TO THEM. WHAT SHOULD YOU CHECK FIRST?

- A. Target Group Health Check Configuration
- B. Security Group Rules
- C. Listener Rule Priority
- D. Auto Scaling Policies



ANSWER: A

Health checks determine if targets are healthy. Incorrect settings can mark unhealthy instances as healthy.

WHY OTHERS ARE WRONG:

- B: Security groups affect connectivity, not health status.
- C: Listener rules control routing, not health.
- D: Auto Scaling affects capacity, not health checks.

4. YOU WANT TO DEPLOY A NEW VERSION OF YOUR APPLICATION WITH 10% TRAFFIC FOR TESTING BEFORE FULL DEPLOYMENT. WHAT IS THE BEST APPROACH?

- A. Create a new Target Group and use weighted routing
- B. Replace all instances and roll back if needed
- C. Change the DNS record
- D. Increase instance size for the new version



ANSWER: B

ALB listener rules with weighted target groups allow traffic shifting for canary deployments.

WHY OTHERS ARE WRONG:

- B: Full replacement causes downtime and risk.
- C: DNS changes are slow and not granular.
- D: Instance size change doesn't shift traffic.

5. YOUR INTERNAL APPLICATION REQUIRES STATIC IP ADDRESSES AND HIGH AVAILABILITY ACROSS AZs. WHICH LOAD BALANCER SHOULD YOU USE?

- A. Application Load Balancer (ALB)
- B. Network Load Balancer (NLB)
- C. Classic Load Balancer (CLB)
- D. Gateway Load Balancer (GWLB)



ANSWER: B

NLB supports static IP addresses (Elastic IP) and provides high availability across AZs.

WHY OTHERS ARE WRONG:

- A: ALB does not support static IPs.
- C: CLB is legacy and not recommended.
- D: GWLB is for appliance insertion, not general traffic.

ELB TRAFFIC FLOW



ELB TYPES QUICK REFERENCE

TYPE	LAYER	BEST FOR	KEY FEATURES
ALB	LAYER 7 (HTTP/HTTPS)	Web apps, microservices	Path/host routing, rules, WAF, auth, redirects
NLB	LAYER 4 (TCP/UDP)	High performance, low latency	Static IPs, millions of requests/sec, TLS pass-through
CLB	LAYER 4/7 (LEGACY)	Basic load balancing	Limited features, legacy
GWLB	LAYER 3 (GRE)	Third-party appliances	Appliance insertion at scale



PRODUCTION TIP

Use health checks, target groups, and traffic shifting together for safe deployments, high availability, and fast recovery.



EXAM STRATEGY

Look for keywords: route, distribute, balance, latency, health, shift traffic, deploy, high availability, static IP.

ECS AND FARGATE

DEVOPS OPERATIONS

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Reliable Deployments
- High Availability
- Efficient Scaling
- Faster Recovery
- Cost Optimization



TASK
DEFINITIONS



SERVICES



DEPLOYMENT
CONFIGURATION



ROLLING
UPDATES



SERVICE
AUTO SCALING



FAILED
DEPLOYMENTS

1. YOUR APPLICATION RUNS ON ECS FARGATE. YOU NEED TO DEFINE THE CPU, MEMORY, PORTS, AND CONTAINER IMAGE FOR YOUR APP. WHICH ECS COMPONENT DOES THIS?

- A. Service
- B. Cluster
- C. Task Definition
- D. Capacity Provider



ANSWER: C

Task Definition specifies the container configuration, resources, networking mode, and IAM roles.

WHY OTHERS ARE WRONG:

- A: Service runs and maintains the desired number of tasks.
- B: Cluster is the logical group of resources.
- D: Capacity Provider manages infrastructure capacity.

2. YOU NEED TO RUN AND MAINTAIN A DESIRED NUMBER OF TASKS AND ENSURE HIGH AVAILABILITY FOR YOUR APPLICATION. WHICH ECS COMPONENT SHOULD YOU USE?

- A. Task Definition
- B. Service
- C. Task Set
- D. Container Instance



ANSWER: B

A Service maintains and scales the desired number of tasks and provides high availability.

WHY OTHERS ARE WRONG:

- A: Task Definition only defines how a container runs.
- C: Task Set is used during deployments, not for steady state.
- D: Container Instance is for EC2 launch type, not Fargate.

3. YOU DEPLOY A NEW VERSION OF YOUR SERVICE. HOW CAN YOU ENSURE ZERO DOWNTIME DURING THE DEPLOYMENT?

- A. Use Rolling Update with minimumHealthyPercent set high
- B. Use Recreate deployment configuration
- C. Scale the service to zero before deployment
- D. Disable Service Auto Scaling



ANSWER: A

Rolling Update replaces tasks gradually while keeping existing tasks running, ensuring zero downtime.

WHY OTHERS ARE WRONG:

- B: Recreate stops all tasks before starting new ones.
- C: Scaling to zero causes full downtime.
- D: Disabling Auto Scaling does not affect deployment strategy.

4. YOUR SERVICE TRAFFIC IS UNPREDICTABLE. YOU NEED TO AUTOMATICALLY SCALE THE NUMBER OF TASKS BASED ON CPU UTILIZATION AROUND 60%. WHAT SHOULD YOU USE?

- A. Step Scaling
- B. Target Tracking Scaling
- C. Manual Scaling
- D. Scheduled Scaling



ANSWER: B

Target Tracking Scaling automatically adjusts tasks to maintain the target CPU utilization.

WHY OTHERS ARE WRONG:

- A: Step Scaling scales in fixed steps, not based on a target.
- C: Manual Scaling requires human intervention.
- D: Scheduled Scaling is time-based, not metric-based.

5. A DEPLOYMENT FAILED AND NEW TASKS CONTINUOUSLY STOP. YOU NEED TO ROLLBACK TO THE PREVIOUS STABLE VERSION AUTOMATICALLY. WHAT IS THE BEST APPROACH?

- A. Use Deployment Circuit Breaker with Rollback enabled
- B. Increase task count and try again
- C. Delete the service and recreate it
- D. Disable health checks



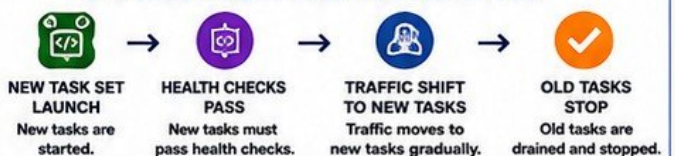
ANSWER: A

Deployment Circuit Breaker detects failures and rolls back to the last stable deployment automatically.

WHY OTHERS ARE WRONG:

- B: Increasing count does not fix the root cause.
- C: Deleting service causes downtime and data loss risk.
- D: Disabling health checks hides issues, doesn't fix them.

ECS DEPLOYMENT ROLLING UPDATE FLOW



ECS COMPONENTS QUICK REFERENCE

COMPONENT	PURPOSE
TASK DEFINITION	Defines container image, CPU, memory, ports, environment variables, IAM roles.
SERVICE	Runs and maintains desired number of tasks, ensures high availability.
DEPLOYMENT CONFIG	Controls how new tasks are deployed (rolling update or recreate).
SERVICE AUTO SCALING	Automatically scales tasks based on metrics or schedule.
HEALTH CHECKS	Ensure tasks are healthy before they receive traffic.



PRODUCTION TIP

Combine healthy deployments, proper scaling policies, and monitoring to ensure resilient and cost-effective ECS workloads.



EXAM STRATEGY

Look for keywords: scale, deployment, health, rollback, traffic, task, service, target, update.

AMAZON EKS

DEVOPS OPERATIONS

5 SCENARIO QUESTIONS + ANSWERS



EXAM FOCUS

- Reliable Kubernetes Operations
- High Availability Clusters
- Efficient Scaling
- Fast Failure Recovery
- Observability & Logging



CLUSTER
DEPLOYMENTS



MANAGED
NODE GROUPS



IAM
ROLES



SCALING
STRATEGIES



LOGGING &
MONITORING



DEPLOYMENT
FAILURES

1. YOU NEED TO DEPLOY A HIGHLY AVAILABLE EKS CLUSTER ACROSS MULTIPLE AZs WITH MINIMAL MANAGEMENT OVERHEAD. WHAT IS THE BEST APPROACH?

- A. Create EC2 instances and install Kubernetes
- B. Use Amazon EKS with Managed Node Groups
- C. Use Fargate only with no node groups
- D. Use EKS Anywhere on on-premises



ANSWER: B

EKS with Managed Node Groups provides high availability across AZs and reduces operational overhead.

WHY OTHERS ARE WRONG:

- A: Manual Kubernetes is not highly available by default.
- C: Fargate alone may not support all workloads and adds limits.
- D: EKS Anywhere is for on-premises or edge environments.

2. YOUR APPLICATION PODS NEED TO ACCESS AWS RESOURCES SECURELY WITHOUT USING LONG-TERM CREDENTIALS. WHAT SHOULD YOU USE?

- A. Store AWS keys in Kubernetes Secrets
- B. Use IAM Roles for Service Accounts (IRSA)
- C. Hardcode IAM user credentials in ConfigMaps
- D. Mount AWS keys using hostPath volumes



ANSWER: B

IRSA allows Kubernetes service accounts to assume IAM roles securely without long-term credentials.

WHY OTHERS ARE WRONG:

- A: Secrets can be compromised and require rotation.
- C: Hardcoding credentials is insecure and not recommended.
- D: HostPath volumes expose sensitive data to the node.

3. YOUR EKS CLUSTER EXPERIENCES HIGH TRAFFIC, AND YOU NEED TO SCALE THE WORKLOAD AUTOMATICALLY BASED ON CPU UTILIZATION. WHAT SHOULD YOU IMPLEMENT?

- A. Manual scaling by editing deployments
- B. Cluster Autoscaler for nodes and HPA for pods
- C. Increase replica count and node count manually
- D. Use only Cluster Autoscaler



ANSWER: B

HPA scales pods based on metrics, and Cluster Autoscaler scales nodes based on resource demand.

WHY OTHERS ARE WRONG:

- A: Manual scaling is slow and error-prone.
- C: Not automatic and requires constant monitoring.
- D: Cluster Autoscaler only scales nodes, not pods.

4. YOUR EKS APPLICATION IS DEPLOYED, BUT NEW PODS ARE STAYING IN PENDING STATE. WHAT IS THE MOST LIKELY CAUSE AND WHAT SHOULD YOU CHECK FIRST?

- A. Insufficient CPU requests in the pod
- B. Insufficient nodes or incorrect node IAM permissions
- C. Incorrect service type configuration
- D. Missing liveness probe



ANSWER: B

Pending pods are most commonly caused by insufficient worker nodes or node role/permissions issues.

WHY OTHERS ARE WRONG:

- A: CPU requests don't stop scheduling if resources exist.
- C: Service type does not affect pod scheduling.
- D: Liveness probe affects running pods, not scheduling.

5. YOUR APPLICATION PODS CRASH AFTER DEPLOYMENT, AND LOGS ARE NOT VISIBLE. WHAT SHOULD YOU DO FIRST TO TROUBLESHOOT?

- A. Check Kubernetes events and pod logs in CloudWatch
- B. Recreate the cluster
- C. Increase node size
- D. Disable network policies



ANSWER: A

Events and logs provide the reason for pod crashes and identify misconfigurations or errors.

WHY OTHERS ARE WRONG:

- B: Recreating the cluster is unnecessary and disruptive.
- C: Larger nodes do not fix application-level issues.
- D: Disabling network policies does not show logs or events.

EKS DEPLOYMENT FLOW



EKS OPERATIONS QUICK REFERENCE

FEATURE	BEST PRACTICE
MANAGED NODE GROUPS	Use MNG for easy scaling, updates, and HA across AZs.
IAM ROLES (IRSA)	Use IRSA for secure access to AWS resources.
SCALING	Use HPA for pods and Cluster Autoscaler for nodes.
LOGGING	Send logs to CloudWatch using Fluent Bit or FireLens.
UPDATES	Use Rolling Updates with maxUnavailable settings.
FAILURE RECOVERY	Use liveness/readiness probes and PDBs.



PRODUCTION TIP

Use managed node groups, automate scaling, implement proper logging, and monitor continuously to run reliable and cost-effective EKS environments.



EXAM STRATEGY

Look for keywords: cluster, deploy, scale, nodes, roles, logs, fail, crash, pending, update.

AWS LAMBDA DEPLOYMENT

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Versions & Aliases
- Provisioned Concurrency
- Canary Deployments
- CodeDeploy Integration
- Rollbacks



VERSIONS

Immutable
code snapshots



ALIASES

Stable endpoints
for versions



PROVISIONED CONCURRENCY

Eliminate cold starts
for critical traffic



CANARY DEPLOYMENTS

Shift traffic safely
with monitoring



CODEDEPLOY INTEGRATION

Automate deployments
& rollbacks

1. A TEAM NEEDS TO DEPLOY NEW CODE WITHOUT AFFECTING CLIENTS UNTIL THEY ARE READY. WHAT SHOULD THEY USE?

- A. Update the \$LATEST version
- B. Create a new version and update the alias
- C. Replace the function directly
- D. Increase the timeout



ANSWER: B

Aliases provide a stable endpoint. Point the alias to a new version to shift traffic without impacting users.

WHY OTHERS ARE WRONG:

- A: \$LATEST changes immediately and is not versioned.
- C: Direct updates cause downtime and risk.
- D: Timeout does not control deployment.

2. A FUNCTION EXPERIENCES HIGH COLD STARTS DURING PEAK HOURS. WHAT IS THE BEST SOLUTION?

- A. Increase memory size
- B. Enable Provisioned Concurrency
- C. Reduce function timeout
- D. Add more environment variables



ANSWER: B

Provisioned Concurrency keeps instances warm and ready, eliminating most cold starts.

WHY OTHERS ARE WRONG:

- A: More memory may help performance, not cold starts.
- C: Timeout does not affect cold starts.
- D: Environment variables do not impact cold starts.

3. YOU WANT TO TEST NEW CODE BY SENDING A SMALL PERCENTAGE OF TRAFFIC TO IT BEFORE FULL DEPLOYMENT. WHAT IS THE BEST APPROACH?

- A. Create a new version and use a canary deployment
- B. Update the function directly
- C. Change the function name
- D. Increase reserved concurrency



ANSWER: A

Canary deployments gradually shift traffic to the new version (e.g., 10% → 100%) while monitoring metrics.

WHY OTHERS ARE WRONG:

- B: Direct update risks all traffic.
- C: New name breaks existing integrations.
- D: Reserved concurrency does not split traffic.

4. YOU NEED AUTOMATED DEPLOYMENTS AND EASY ROLLBACKS FOR LAMBDA. WHAT SHOULD YOU INTEGRATE?

- A. AWS CodeDeploy
- B. AWS CloudFormation only
- C. AWS Step Functions
- D. Amazon EventBridge



ANSWER: A

CodeDeploy supports Lambda deployments, canary/linear shifts, automatic rollback on alarm failures.

WHY OTHERS ARE WRONG:

- B: CloudFormation deploys but lacks traffic shifting.
- C: Step Functions orchestrates workflows, not deployments.
- D: EventBridge triggers events, not deployments.

5. A DEPLOYMENT CAUSED ERRORS AFTER TRAFFIC SHIFT. HOW CAN YOU ROLLBACK QUICKLY?

- A. Point the alias back to the previous version
- B. Delete the function and recreate it
- C. Disable CloudWatch alarms
- D. Increase memory and retry



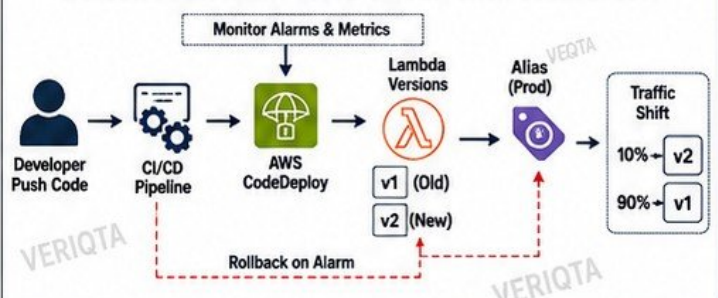
ANSWER: A

Re-point the alias to the last known good version instantly restores stable traffic.

WHY OTHERS ARE WRONG:

- B: Recreating causes downtime and breaks integrations.
- C: Disabling alarms hides issues.
- D: More memory does not fix bad code.

LAMBDA DEPLOYMENT FLOW (CANARY WITH CODEDEPLOY)



PRODUCTION TIP

Always use aliases for production. Never invoke versions directly.



BEST PRACTICE

Combine Provisioned Concurrency with canary deployments for zero cold starts and safe releases.



EXAM STRATEGY

Look for keywords: "zero downtime", "gradual traffic", "automatic rollback", "stable endpoint" → think Aliases + CodeDeploy.

SERVERLESS CI/CD PIPELINE

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Lambda
- API Gateway
- SAM
- CodePipeline
- Deployment Preferences
- Automated Testing



LAMBDA
Serverless compute



API GATEWAY
Expose APIs securely



AWS SAM
Serverless app modeling



CODEPIPELINE
Automate build, test & deploy



DEPLOYMENT PREFERENCES
Safe & controlled releases



AUTOMATED TESTING
Validate every change

1. A TEAM NEEDS A CI/CD PIPELINE TO DEPLOY A LAMBDA FUNCTION AND API GATEWAY USING INFRASTRUCTURE AS CODE. WHICH COMBINATION IS THE BEST CHOICE?

- A. CodeBuild + S3 + Manual deployment
- B. CodePipeline + AWS SAM + CloudFormation
- C. CodeCommit + EC2 + Ansible
- D. CodeDeploy + Elastic Beanstalk



ANSWER: B

AWS SAM generates CloudFormation templates. CodePipeline automates build, test, and deploy of serverless stacks.

WHY OTHERS ARE WRONG:

- A: No automation and not serverless-native.
- C: EC2 and Ansible are not needed for serverless.
- D: CodeDeploy does not deploy SAM templates directly.

2. YOU DEPLOY A NEW VERSION OF A LAMBDA FUNCTION BUT SEE ERRORS AFTER DEPLOYMENT. HOW CAN YOU ROLLBACK QUICKLY WITH MINIMAL IMPACT?

- A. Update the function code and redeploy
- B. Use an alias and shift traffic back to the previous version
- C. Delete the function and recreate it
- D. Disable the function in API Gateway



ANSWER: B

Aliases allow you to shift traffic between versions instantly, enabling fast rollback.

WHY OTHERS ARE WRONG:

- A: Redeploying takes time and risk.
- C: Deleting causes downtime and breaks integrations.
- D: Disabling API Gateway impacts all versions.

3. A TEAM WANTS TO DEPLOY NEW FEATURES TO API GATEWAY AND LAMBDA WITH ZERO DOWNTIME AND EASY ROLLBACK. WHAT IS THE BEST APPROACH?

- A. Deploy directly to \$latest version
- B. Use alias + CodeDeploy with linear or canary deployment
- C. Update environment variables manually
- D. Increase timeout and retry



ANSWER: B

CodeDeploy with aliases enables traffic shifting (canary/linear) for safe deployments and instant rollback.

WHY OTHERS ARE WRONG:

- A: \$latest is not stable and cannot be rolled back.
- C: Env vars do not control deployment strategy.
- D: Timeout/retry do not prevent bad releases.

4. YOUR SERVERLESS PIPELINE NEEDS AUTOMATED TESTING BEFORE DEPLOYMENT. WHERE SHOULD TESTS BE ADDED FOR BEST FEEDBACK?

- A. Only after deployment
- B. In CodePipeline after build but before deploy
- C. Manually by developers
- D. In CloudFormation



ANSWER: B

Automated tests in the pipeline catch issues early, preventing bad deployments.

WHY OTHERS ARE WRONG:

- A: Testing after deploy risks impacting users.
- C: Manual testing is slow and inconsistent.
- D: CloudFormation does not run application tests.

5. A PIPELINE DEPLOYS MULTIPLE SERVERLESS FUNCTIONS. ONE FUNCTION FAILS IN SAM DEPLOYMENT. WHAT IS THE BEST WAY TO DEBUG AND FIX THE ISSUE?

- A. Check CloudWatch Logs only
- B. Check SAM build logs and CloudFormation events
- C. Delete the stack and redeploy
- D. Restart the pipeline



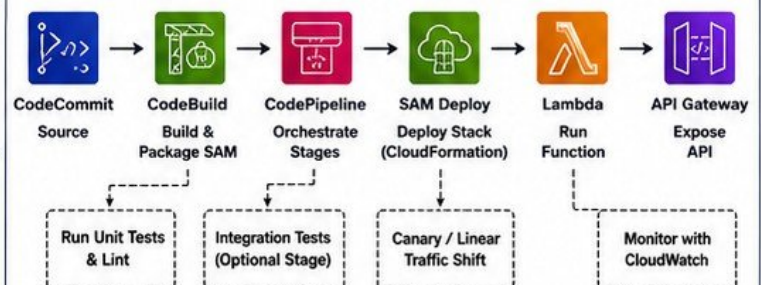
ANSWER: B

SAM build logs show packaging/build issues, and CloudFormation events show deployment errors.

WHY OTHERS ARE WRONG:

- A: Logs show runtime issues, not deployment failures.
- C: Deleting causes downtime and data loss risk.
- D: Restarting won't fix the underlying issue.

SERVERLESS CI/CD PIPELINE FLOW



PRODUCTION TIP

Always use aliases for Lambda. Never invoke versions directly. Aliases give you safety.



BEST PRACTICE

Use SAM for consistent, repeatable serverless deployments across environments.



EXAM STRATEGY

Look for keywords: "zero downtime", "rollback", "traffic shift", "automation", "safe deployment" → Alias + CodeDeploy.

HIGH AVAILABILITY AND DISASTER RECOVERY

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Architecture choices
- Recovery strategies
- RTO & RPO understanding
- Cost vs. resilience
- Best solution selection



MULTI-AZ
Protect within
one Region



MULTI-REGION
Protect across
Regions



BACKUP & RESTORE
Recover data
quickly



PILOT LIGHT
Minimal resources
in standby



WARM STANDBY
Scaled-down
environment



**RECOVERY
OBJECTIVES**
RTO, RPO, SLA

1. YOUR APPLICATION IS RUNNING ON EC2 INSTANCES AND AN RDS DATABASE. USERS IN ONE AZ ARE EXPERIENCING OUTAGES DUE TO AZ FAILURE. WHAT SHOULD YOU DO TO IMPROVE AVAILABILITY?

- A. Deploy resources across multiple AZs
- B. Enable Multi-Region active-active
- C. Use S3 for all data
- D. Increase instance size



ANSWER: A

Using multiple AZs keeps your application running if one Availability Zone fails.

WHY OTHERS ARE WRONG:

- B: Multi-Region is for region-level disasters, not AZ failures.
- C: S3 is not a replacement for application databases.
- D: Bigger instances do not protect against AZ failure.

2. YOUR BUSINESS REQUIRES RTO OF 15 MINUTES AND RPO OF 5 MINUTES. WHICH ARCHITECTURE BEST MEETS THESE REQUIREMENTS?

- A. Multi-AZ within a single Region
- B. Backup to S3 once per day
- C. Multi-Region with automated failover
- D. Pilot light in a secondary Region



ANSWER: C

Multi-Region with automated failover meets aggressive RTO (15 min) and RPO (5 min) requirements.

WHY OTHERS ARE WRONG:

- A: Multi-AZ does not protect against Region-wide failures.
- B: Daily backups cannot meet the RPO requirement.
- D: Pilot light typically has longer recovery time.

3. YOU NEED A COST-EFFECTIVE DR STRATEGY FOR AN APPLICATION WITH RTO OF 2 HOURS AND RPO OF 1 HOUR. WHAT IS THE BEST CHOICE?

- A. Pilot light
- B. Warm standby
- C. Multi-Region active-active
- D. Synchronous replication



ANSWER: A

Pilot light keeps minimal resources running, reducing costs while restoring within the 2-hour RTO.

WHY OTHERS ARE WRONG:

- B: Warm standby is more expensive than pilot light.
- C: Active-active is costly and unnecessary for 2-hour RTO.
- D: Synchronous replication increases cost and complexity.

4. YOUR DATABASE IS CRITICAL AND YOU NEED POINT-IN-TIME RECOVERY AFTER ACCIDENTAL DELETION. WHAT SHOULD YOU IMPLEMENT?

- A. Enable automated backups with point-in-time restore
- B. Increase instance storage
- C. Use an AMI backup
- D. Store data in EBS snapshots only



ANSWER: A

Automated backups with point-in-time restore allow recovery to any moment, minimizing data loss.

WHY OTHERS ARE WRONG:

- B: More storage does not help recover deleted data.
- C: AMI backups do not capture database transaction logs.
- D: Snapshots alone are not enough for point-in-time recovery.

5. A REGION-WIDE OUTAGE OCCURS. HOW CAN YOU QUICKLY RESTORE YOUR APPLICATION WITH MINIMAL DOWNTIME?

- A. Multi-AZ deployment
- B. Multi-Region with Route 53 failover
- C. Increase CloudWatch alarms
- D. Store logs in S3



ANSWER: B

Multi-Region with DNS failover (Route 53) enables quick traffic switching to the healthy Region.

WHY OTHERS ARE WRONG:

- A: Multi-AZ does not protect against Region failures.
- C: Alarms detect issues but do not restore service.
- D: Logs in S3 do not help restore the application.

DISASTER RECOVERY STRATEGIES COMPARISON

STRATEGY	RTO	RPO	COST	USE CASE
PILOT LIGHT	HOURS	HOURS	LOW	Cost-sensitive, tolerates longer downtime
WARM STANDBY	MINUTES	MINUTES	MEDIUM	Balanced cost and fast recovery
MULTI-AZ	MINUTES	MINUTES	LOW	Protects against AZ failures
BACKUP & RESTORE	HOURS+	HOURS+	LOW	Non-critical workloads
MULTI-REGION ACTIVE-ACTIVE	SECONDS	NEAR ZERO	HIGH	Mission-critical global apps



PRODUCTION TIP

Always define RTO and RPO with business teams before designing your DR architecture.



BEST PRACTICE

Test your DR plan regularly. A plan that is not tested is a plan that will fail.



EXAM STRATEGY

Match the recovery objectives to the strategy that meets them with the lowest possible cost.

BACKUP AND RECOVERY AUTOMATION

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- AWS Backup
- Snapshots
- Cross-Region Copies
- Backup Policies
- Restore Testing
- Recovery Automation



AWS BACKUP
Centralized backup service



SNAPSHOTS
Point-in-time recovery



CROSS-REGION COPIES
Protect across Regions



BACKUP POLICIES
Automate schedules & retention



RESTORE TESTING
Validate backups regularly



RECOVERY AUTOMATION
Automate recovery workflows

1. YOU NEED A CENTRALIZED BACKUP SOLUTION FOR MULTIPLE AWS SERVICES WITH DIFFERENT RETENTION PERIODS AND AUTOMATED SCHEDULING. WHAT IS THE BEST APPROACH?

- A. Create snapshots manually for each service
- B. Use AWS Backup with a backup policy
- C. Use AWS DataSync
- D. Use CloudTrail for backups



ANSWER: B

AWS Backup provides a centralized service to configure backup policies, automate schedules, and manage backups across many AWS services.

WHY OTHERS ARE WRONG:

- A: Manual snapshots do not scale and are error-prone.
- C: DataSync is for data transfer, not backups.
- D: CloudTrail records API activity, not backups.

2. YOUR APPLICATION STORES DATA ON AN EBS VOLUME IN US-EAST-1. YOU NEED TO PROTECT AGAINST A REGION-WIDE DISASTER. WHAT SHOULD YOU DO?

- A. Take daily snapshots only in us-east-1
- B. Create cross-Region copies of the snapshots to us-west-2
- C. Increase the volume size for redundancy
- D. Enable S3 versioning



ANSWER: B

Cross-Region copy of EBS snapshots ensures data is available in another Region if the primary Region fails.

WHY OTHERS ARE WRONG:

- A: Snapshots in one Region do not protect against Region failure.
- C: Larger volume does not provide disaster recovery.
- D: S3 versioning is for objects in S3, not EBS volumes.

3. YOU NEED TO ENSURE THAT BACKUPS ARE AUTOMATICALLY CREATED DAILY AT 2 AM AND RETAINED FOR 35 DAYS FOR EC2 INSTANCES AND RDS DATABASES. WHAT SHOULD YOU USE?

- A. AWS Backup backup plan with lifecycle rules
- B. CloudWatch Events rule
- C. AWS Config rule
- D. Manual snapshot script



ANSWER: A

AWS Backup backup plans allow you to schedule backups and set retention (lifecycle) rules for resources.

WHY OTHERS ARE WRONG:

- B: CloudWatch Events can trigger actions but does not manage backup retention.
- C: AWS Config monitors resources, not manage backups.
- D: Manual scripts do not scale and lack lifecycle management.

4. YOU NEED TO TEST YOUR BACKUPS REGULARLY TO ENSURE THEY CAN BE RESTORED SUCCESSFULLY. WHAT IS THE BEST WAY TO ACHIEVE THIS?

- A. Assume backups always work until needed
- B. Restore backups in a lower environment on a schedule
- C. Store backups in cold storage only
- D. Increase retention period



ANSWER: B

Restore backups to a test environment on a schedule to verify integrity and recovery procedures.

WHY OTHERS ARE WRONG:

- A: This is risky and not a best practice.
- C: Cold storage does not validate recoverability.
- D: Retention is not a test: longer retention ≠ valid backups.

5. AFTER A RANSOMWARE ATTACK, YOU NEED TO AUTOMATICALLY RECOVER YOUR APPLICATION USING INFRASTRUCTURE AS CODE AND LATEST BACKUPS. WHAT IS THE BEST APPROACH?

- A. Restore resources manually from the console
- B. Use AWS Backup with Recovery Automation and CloudFormation
- C. Copy data from S3 manually
- D. Rebuild everything from scratch



ANSWER: B

Recovery Automation can orchestrate restore steps using CloudFormation, backups, and runbooks to recover quickly.

WHY OTHERS ARE WRONG:

- A: Manual recovery is slow and error-prone.
- C: S3 copy does not recover full application stack.
- D: Rebuilding everything is time-consuming.

BACKUP AND RECOVERY AUTOMATION WORKFLOW



PRODUCTION TIP

Follow the 3-2-1-1-0 rule: 3 copies, 2 different media, 1 offsite copy, 1 offline copy, 0 errors (test restores).



BEST PRACTICE

Automate backups, replicate across Regions, test restores, and automate recovery to meet RTO and RPO.



EXAM STRATEGY

Match the requirement to the service: schedule & retention → AWS Backup, Region protection → Cross-Region, testing → restore in test environment.

MULTI-ACCOUNT DEVOPS ARCHITECTURE

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- AWS Organizations
- OUs and Account Structure
- SCPs and Guardrails
- Shared Services Strategy
- Centralized Logging
- Deployment Best Practices



AWS ORGANIZATIONS
Centrally manage multiple accounts



ORGANIZATIONAL UNITS
Group accounts logically



SCPs
Apply guardrails using policies



SHARED SERVICES
Provide common capabilities



CENTRALIZED LOGGING
Collect logs in a central account



DEPLOYMENT ACCOUNTS
Run workloads and applications

1. YOU NEED TO PREVENT DEVELOPERS FROM LAUNCHING EC2 INSTANCES IN THE PRODUCTION ACCOUNTS. WHAT IS THE BEST WAY TO ENFORCE THIS?

- A. Use IAM policies in each account
- B. Use an SCP to deny EC2:RunInstances
- C. Use CloudWatch alarms
- D. Use Security Groups

✓ ANSWER: B

SCPs apply guardrails at the organization level and prevent the action in all accounts within the scope.

WHY OTHERS ARE WRONG:

- A: IAM policies can be bypassed by admins with more permissions.
- C: Alarms detect but do not prevent the action.
- D: Security Groups do not control creation of resources.

2. YOU WANT TO CENTRALIZE CLOUDTRAIL LOGS FROM ALL ACCOUNTS FOR AUDIT AND COMPLIANCE PURPOSES. WHAT IS THE BEST APPROACH?

- A. Enable CloudTrail in each account and send logs to S3 in a central logging account
- B. Use AWS Config in the management account
- C. Use CloudWatch cross-account dashboards
- D. Use S3 replication between all accounts

✓ ANSWER: A

Deliver CloudTrail logs from all accounts to a central S3 bucket for auditing and compliance.

WHY OTHERS ARE WRONG:

- B: AWS Config does not collect CloudTrail logs.
- C: Dashboards display but do not centralize logs.
- D: S3 replication is not reliable for log aggregation.

3. YOU NEED TO ORGANIZE ACCOUNTS INTO DIFFERENT TEAMS AND ENVIRONMENTS WITH SEPARATE BUDGETS AND POLICIES. WHAT SHOULD YOU USE?

- A. Tags
- B. Organizational Units (OUs)
- C. IAM Groups
- D. VPCs

✓ ANSWER: B

OUs allow you to group accounts logically and apply policies, budgets, and controls to the group.

WHY OTHERS ARE WRONG:

- A: Tags do not provide account-level governance.
- C: IAM Groups manage users, not accounts.
- D: VPCs are networking constructs, not account containers.

4. YOU NEED A CENTRAL LOCATION TO MANAGE CI/CD TOOLS, ARTIFACT REPOSITORIES, AND SHARED NETWORK SERVICES FOR ALL ACCOUNTS. WHAT IS THE BEST DESIGN?

- A. Use the management account
- B. Create a dedicated shared services account
- C. Deploy in every account
- D. Use a single region for all workloads

✓ ANSWER: B

A shared services account centralizes common services like CI/CD, artifacts, monitoring, and networking.

WHY OTHERS ARE WRONG:

- A: Management account is not meant for workloads.
- C: Duplicates effort and increases cost.
- D: Single region reduces resilience and availability.

5. YOU NEED TO DEPLOY APPLICATIONS TO DIFFERENT ACCOUNTS BASED ON ENVIRONMENT (DEV, TEST, PROD) WITH ISOLATED PERMISSIONS. WHAT IS THE BEST PRACTICE?

- A. Deploy all environments in one account
- B. Use separate deployment accounts for each environment
- C. Use IAM roles only
- D. Use different regions only

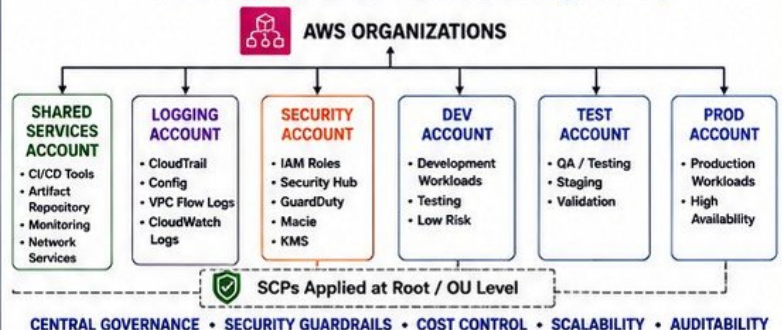
✓ ANSWER: B

Separate accounts provide strong isolation, security, blast-radius reduction, and independent scaling.

WHY OTHERS ARE WRONG:

- A: Increases risk of accidental changes.
- C: Roles do not provide account-level isolation.
- D: Regions alone do not provide strong isolation.

MULTI-ACCOUNT DEVOPS ARCHITECTURE (EXAMPLE)



PRODUCTION TIP

Design your account structure early. Changing it later is complex and risky.



BEST PRACTICE

Use least-privilege access and SCPs to enforce standards across all accounts.



EXAM STRATEGY

Think in terms of governance, security, isolation, and repeatable deployments across multiple accounts.

GOVERNANCE AND COMPLIANCE

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- AWS Config
- CloudTrail
- Organizations
- Control Tower
- Security Hub
- Compliance Automation



AWS CONFIG
Track & evaluate resource compliance



CLOUDTRAIL
Audit API activity & changes



ORGANIZATIONS
Centralized governance at scale



CONTROL TOWER
Set up & govern secure multi-account landing zone



SECURITY HUB
Central view of security findings & compliance



COMPLIANCE AUTOMATION
Automate checks, remediation & reporting

1. YOU NEED TO ENSURE THAT S3 BUCKETS ARE ENCRYPTED AND PUBLIC ACCESS IS BLOCKED ACROSS ALL ACCOUNTS. WHAT IS THE MOST EFFECTIVE APPROACH?

- A. Use IAM policies in each account
- B. Use AWS Config rules with automatic remediation
- C. Use Security Groups
- D. Use CloudWatch alarms



ANSWER: B

AWS Config rules can detect non-compliant S3 buckets and automatically remediate them.

WHY OTHERS ARE WRONG:

- A: IAM policies do not evaluate existing resource settings.
- C: Security Groups do not apply to S3.
- D: CloudWatch alarms only notify; they do not remediate.

2. YOU NEED A COMPLETE AUDIT TRAIL OF ALL API ACTIVITY FOR COMPLIANCE INVESTIGATIONS. WHAT SHOULD YOU DO?

- A. Enable CloudTrail organization trail with all regions
- B. Enable AWS Config in each account
- C. Use VPC Flow Logs
- D. Use GuardDuty only



ANSWER: A

CloudTrail organization trail captures API activity across all accounts and regions.

WHY OTHERS ARE WRONG:

- B: Config records configuration changes, not API calls.
- C: VPC Flow Logs record network traffic, not API activity.
- D: GuardDuty detects threats; it does not provide audit logs.

3. YOU NEED TO PREVENT THE CREATION OF IAM USERS OUTSIDE OF THE IDENTITY ACCOUNT. WHAT SHOULD YOU USE?

- A. IAM permissions boundary
- B. AWS Organizations SCP
- C. CloudWatch metric filter
- D. Security Hub control



ANSWER: B

Service Control Policies (SCPs) can prevent IAM user creation in non-approved accounts.

WHY OTHERS ARE WRONG:

- A: Permissions boundaries restrict actions within an account.
- C: CloudWatch only monitors; it does not prevent actions.
- D: Security Hub reports findings but does not enforce.

4. YOU NEED A CENTRAL DASHBOARD TO MONITOR COMPLIANCE STATUS, SECURITY FINDINGS, AND BEST PRACTICES ACROSS ALL ACCOUNTS. WHAT SHOULD YOU USE?

- A. AWS Config dashboards only
- B. AWS Security Hub
- C. CloudTrail Insights
- D. AWS Cost Explorer



ANSWER: B

Security Hub aggregates findings and compliance status from multiple AWS services in a single dashboard.

WHY OTHERS ARE WRONG:

- A: Config shows compliance but not security findings.
- C: CloudTrail Insights detects anomalies, not compliance.
- D: Cost Explorer is for cost analysis, not compliance.

5. YOU NEED TO AUTOMATE COMPLIANCE REPORTING FOR PCI-DSS AND ENSURE EVIDENCE IS COLLECTED REGULARLY. WHAT IS THE BEST APPROACH?

- A. Use manual scripts to collect logs
- B. Use AWS Config conformance packs with SSM Automation
- C. Use CloudWatch dashboards
- D. Use Trusted Advisor reports



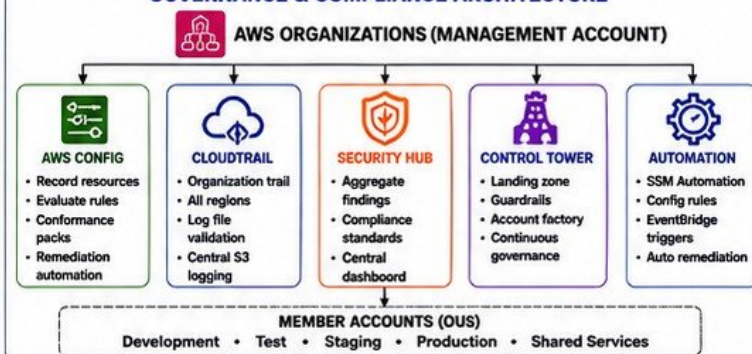
ANSWER: B

Config conformance packs evaluate compliance continuously, while SSM Automation can collect and store evidence.

WHY OTHERS ARE WRONG:

- A: Manual scripts are error-prone and not scalable.
- C: CloudWatch dashboards do not evaluate compliance.
- D: Trusted Advisor provides suggestions, not compliance checks.

GOVERNANCE & COMPLIANCE ARCHITECTURE



PRODUCTION TIP

Automate governance early. Manual compliance does not scale and increases risk.



BEST PRACTICE

Use Organizations + SCPs for prevention and Config + Automation for continuous compliance.



EXAM STRATEGY

Match the requirement to the service: detect → Config, audit → CloudTrail, enforce → SCPs, monitor → Security Hub, automate → SSM Automation.

PERFORMANCE AND COST OPTIMIZATION

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Auto Scaling
- Compute selection
- CloudWatch metrics
- Cost allocation
- Resource rightsizing
- Automated optimization



AUTO SCALING
Scale based on demand



COMPUTE SELECTION
Right instance for the job



CLOUDWATCH METRICS
Monitor, alert, optimize



COST ALLOCATION
Track costs by team, app, or environment



RESOURCE RIGHTSIZING
Right size to reduce waste



AUTOMATED OPTIMIZATION
Use tools to optimize cost

1. YOUR APPLICATION TRAFFIC IS UNPREDICTABLE AND INSTANCES ARE OFTEN UNDERUTILIZED AT NIGHT. WHAT IS THE MOST COST-EFFECTIVE SOLUTION?

- A. Increase instance size
- B. Implement Auto Scaling with target tracking
- C. Use Reserved Instances for maximum savings
- D. Migrate to a single larger instance



ANSWER: B

Auto Scaling with target tracking scales in and out based on demand, ensuring optimal performance and cost.

WHY OTHERS ARE WRONG:

- A: Bigger instances cost more even when idle.
- C: Reserved Instances save cost but do not handle variability.
- D: Single instance is not resilient and wastes resources.

2. YOU NEED TO CHOOSE THE MOST COST-EFFECTIVE OPTION TO RUN A CONTAINERIZED WORKLOAD THAT HAS STEADY, PREDICTABLE TRAFFIC.

- A. Amazon EC2 On-Demand Instances
- B. AWS Fargate
- C. Amazon EC2 Spot Instances
- D. AWS Lambda



ANSWER: C

Spot Instances provide significant savings for steady workloads that can tolerate interruption.

WHY OTHERS ARE WRONG:

- A: On-Demand is the most expensive option.
- B: Fargate is convenient but costs more for steady workloads.
- D: Lambda is for event-driven, not long-running containers.

3. CLOUDWATCH SHOWS HIGH CPU UTILIZATION (90%+) FOR YOUR INSTANCES OVER THE LAST 3 DAYS. WHAT SHOULD YOU DO FIRST?

- A. Increase the alarm threshold
- B. Investigate and optimize the workload
- C. Scale down instances
- D. Stop instances during peak hours



ANSWER: B

High CPU indicates a performance issue; investigate and optimize before scaling.

WHY OTHERS ARE WRONG:

- A: Hiding the problem does not resolve it.
- C: Scaling down will worsen performance.
- D: Stopping during peak hours causes outages.

4. YOUR COMPANY WANTS TO TRACK AND REPORT COSTS BY TEAM AND APPLICATION FOR BUDGETING AND CHARGEBACK. WHAT IS THE BEST APPROACH?

- A. Use the Billing console only
- B. Implement AWS Cost Allocation Tags
- C. Use CloudWatch metrics
- D. Enable trusted advisor reports



ANSWER: B

Cost Allocation Tags let you track and allocate costs by team, app, or environment.

WHY OTHERS ARE WRONG:

- A: Billing console shows costs but does not allocate.
- C: CloudWatch does not provide cost allocation.
- D: Trusted Advisor provides recommendations, not allocation.

5. YOU WANT TO CONTINUOUSLY OPTIMIZE COSTS AND IDENTIFY RIGHTSIZING OPPORTUNITIES AUTOMATICALLY. WHAT IS THE BEST COMBINATION?

- A. Trusted Advisor + AWS Compute Optimizer
- B. CloudTrail + AWS Config
- C. AWS Backup + CloudWatch
- D. IAM Access Analyzer + GuardDuty



ANSWER: A

Trusted Advisor identifies cost best practices and Compute Optimizer recommends rightsizing opportunities.

WHY OTHERS ARE WRONG:

- B: These services are for governance, not cost optimization.
- C: Related to backup and monitoring, not optimization.
- D: Security-focused services, not cost optimization.

COST OPTIMIZATION FRAMEWORK



1. MONITOR

- CloudWatch
- Metrics
- Alarms
- Dashboards



2. ANALYZE

- Identify trends
- Find waste
- Review costs by tags



3. OPTIMIZE

- Rightsize resources
- Choose right pricing model



4. AUTOMATE

- Auto Scaling
- Schedules
- Reserved/Savings Plans



5. VALIDATE

- Measure results
- Track savings
- Continuously improve



PRODUCTION TIP

Monitor, analyze, optimize, automate, and validate. Repeat continuously.



BEST PRACTICE

Use data, not guesses. Rightsize, automate scaling, and review costs weekly.



EXAM STRATEGY

Look for keywords: "cost-effective", "optimize", "least operational effort", "most efficient" → Consider scaling, rightsizing, and automation.

PRODUCTION TROUBLESHOOTING

5 SCENARIO QUESTIONS + ANSWERS

EXAM FOCUS

- Failed Deployments
- IAM Permission Failures
- Unhealthy Targets
- Pipeline Failures
- CloudFormation Failures
- Missing Logs



FAILED DEPLOYMENTS

Deployment rollbacks & fixes



IAM FAILURES

Identify & fix permission issues



UNHEALTHY TARGETS

Find & recover unhealthy targets



PIPELINE FAILURES

Debug CI/CD pipeline issues



CLOUDFORMATION

Resolve stack failures



MISSING LOGS

Restore visibility & logging

1. YOUR APPLICATION DEPLOYMENT TO ECS KEEPS FAILING AND ROLLS BACK. HOW DO YOU TROUBLESHOOT?

- Increase task count
- Check task definition and container logs
- Delete the service and recreate
- Disable the deployment circuit breaker

✓ ANSWER: B

Check task definition, recent changes, and container logs to identify the failure.

WHY OTHERS ARE WRONG:

- A: More tasks won't fix a bad deployment.
- C: Deleting loses history and does not solve the issue.
- D: Disabling circuit breaker hides failures.

2. USERS ARE GETTING ACCESS DENIED ERRORS WHEN RUNNING A DEPLOYMENT PIPELINE. WHAT IS THE MOST LIKELY CAUSE?

- S3 bucket policy is blocking access
- IAM role used by the pipeline lacks permissions
- VPC endpoint is not available
- CloudTrail is disabled

✓ ANSWER: B

The pipeline's IAM role must have all required permissions to access resources.

WHY OTHERS ARE WRONG:

- A: Bucket policy would show specific S3 denial.
- C: VPC endpoint issues cause timeouts, not AccessDenied.
- D: CloudTrail being disabled does not cause access denial.

3. AN APPLICATION BEHIND AN ALB IS RETURNING 5XX ERRORS. TARGET GROUP HEALTH CHECKS SHOW UNHEALTHY. WHAT SHOULD YOU CHECK FIRST?

- Increase ALB capacity
- Check the application logs on the instances
- Change the load balancer type
- Modify the security group to allow all traffic

✓ ANSWER: B

Application or instance issues are the most common cause of unhealthy targets.

WHY OTHERS ARE WRONG:

- A: ALB is managed and auto-scales.
- C: Changing ALB type won't fix failing targets.
- D: Allowing all traffic is risky and not the root cause.

4. YOUR CODEPIPELINE FAILED AT THE DEPLOY STAGE. WHAT IS THE BEST WAY TO FIND THE ROOT CAUSE?

- Re-run the pipeline immediately
- Check the CloudWatch Logs for the failed stage
- Delete and recreate the pipeline
- Manually deploy the code to production

✓ ANSWER: B

CloudWatch Logs provide detailed error messages and stack traces for the failed action.

WHY OTHERS ARE WRONG:

- A: Re-running may fail again without fixing the cause.
- C: Deleting removes history and does not solve the issue.
- D: Manual deploy bypasses the real issue and is risky.

5. A CLOUDFORMATION STACK IS IN UPDATE_ROLLBACK_FAILED STATE. WHAT IS THE BEST NEXT STEP?

- Delete the stack
- Fix the issues and continue rollback
- Use ContinueUpdateRollback to recover
- Create a new stack with the same template

✓ ANSWER: C

ContinueUpdateRollback helps recover the stack to a stable state after a failed rollback.

WHY OTHERS ARE WRONG:

- A: Deleting may cause data loss.
- B: You cannot fix and rollback automatically.
- D: Creating new stack is not a recovery solution.

TROUBLESHOOTING FLOW



1. IDENTIFY THE ISSUE

- What is failing?
- When did it start?
- Who is affected?



2. GATHER EVIDENCE

- Check logs
- Metrics & alarms
- Recent changes



3. ANALYZE ROOT CAUSE

- Compare expected vs actual
- Isolate the problem



4. RESOLVE AND FIX

- Apply fix
- Validate the fix
- Monitor closely



5. PREVENT RECURRENCE

- Update alerts
- Improve runbooks
- Document lessons



PRODUCTION TIP

Stay calm, gather data, and fix the cause, not just the symptom. Document and improve.



BEST PRACTICE

Use runbooks, automate detection, and monitor everything. Visibility prevents outages.



EXAM STRATEGY

Read carefully, eliminate risky options, and choose the method that provides visibility and long-term reliability.

COMPLEX ARCHITECTURE SCENARIOS

5 ADVANCED QUESTIONS + ANSWERS

EXAM FOCUS

- Multi-account architecture
- Multi-Region systems
- CI/CD
- Security
- Observability
- Automated recovery



MULTI-ACCOUNT ARCHITECTURE
Scalable, secure account structure



MULTI-REGION SYSTEMS
High availability & resilience



CI/CD
Reliable delivery at scale



SECURITY
Defense in depth & compliance



OBSERVABILITY
Logs, metrics, traces, alerts



AUTOMATED RECOVERY
Detect, recover, restore automatically

1. YOU ARE DESIGNING A MULTI-ACCOUNT AWS ENVIRONMENT FOR A LARGE ORGANIZATION WITH STRICT SECURITY REQUIREMENTS. WHAT IS THE BEST APPROACH?

- Use a single account with multiple VPCs
- Use AWS Organizations with a well-structured OU hierarchy
- Share resources across accounts using IAM roles only
- Use resource policies to control access across all accounts



ANSWER: B

AWS Organizations with OUs provides governance, isolation, and centralized control across accounts.

WHY OTHERS ARE WRONG:

- A: Single account increases blast radius and limits isolation.
- C: IAM roles alone do not provide strong governance.
- D: Resource policies are not a full governance solution.

3. YOUR CI/CD PIPELINE DEPLOYS TO MULTIPLE ACCOUNTS AND REGIONS. HOW DO YOU ENSURE SAFE, REPEATABLE, AND AUDITABLE DEPLOYMENTS?

- Manual deployments via SSH
- Centralized CodePipeline + cross-account roles + approvals
- Separate pipelines per developer
- Store credentials in environment variables



ANSWER: B

Centralized CodePipeline with cross-account roles and approvals ensures consistency, security, and auditability.

WHY OTHERS ARE WRONG:

- A: Manual deployments are error-prone and unscalable.
- C: Many pipelines increase cost and maintenance.
- D: Storing credentials insecurely is a major risk.

5. A CRITICAL SERVICE IS DOWN DUE TO AN UNKNOWN ERROR. YOU NEED TO MINIMIZE DOWNTIME AND RECOVER AUTOMATICALLY AS MUCH AS POSSIBLE. WHAT IS THE BEST STRATEGY?

- Restart instances manually and hope it recovers
- Implement monitoring, auto-remediation, and failover
- Increase instance size to handle the load
- Disable alarms to avoid noise



ANSWER: B

Monitoring with automated remediation and failover minimizes MTTR and recovers service automatically when possible.

WHY OTHERS ARE WRONG:

- A: Manual actions are slow and increase downtime.
- C: Bigger instances do not fix unknown errors.
- D: Disabling alarms hides issues and increases risk.

2. YOUR APPLICATION MUST BE HIGHLY AVAILABLE ACROSS MULTIPLE REGIONS WITH LOW LATENCY FOR GLOBAL USERS. WHAT IS THE BEST ARCHITECTURE?

- Single-Region with Auto Scaling
- Multi-Region active-active with Route 53 latency routing
- Multi-AZ in a single Region only
- Warm standby in a secondary Region



ANSWER: B

Active-active across Regions with Route 53 latency routing provides the best availability and performance.

WHY OTHERS ARE WRONG:

- A: Single-Region is not resilient to regional failure.
- C: Multi-AZ does not protect against Region failure.
- D: Warm standby increases recovery time objectives (RTO).

4. YOUR SECURITY TEAM REQUIRES CONTINUOUS COMPLIANCE MONITORING ACROSS ALL ACCOUNTS WITH AUTOMATED REMEDIATION. WHAT IS THE BEST SOLUTION?

- AWS Config + Security Hub + Organizations
- Trusted Advisor only
- IAM Access Analyzer only
- Manual audits and spreadsheets



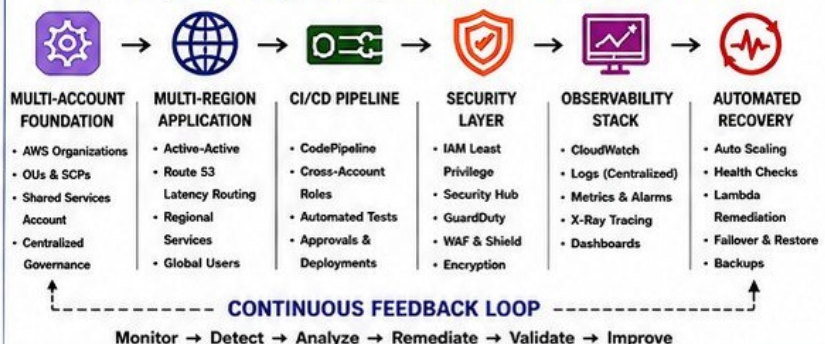
ANSWER: A

AWS Config for rules, Security Hub for findings aggregation, and Organizations for coverage across all accounts.

WHY OTHERS ARE WRONG:

- B: Trusted Advisor provides recommendations, not enforcement.
- C: IAM Access Analyzer focuses only on IAM risks.
- D: Manual audits are slow, error-prone, and not continuous.

COMPLEX, RESILIENT, SECURE, AND OBSERVABLE ARCHITECTURE



PRODUCTION TIP

Design for failure, automate recovery, and test regularly. Practice game days and chaos engineering.



BEST PRACTICE

Follow the Well-Architected Framework: Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization.



EXAM STRATEGY

Think end-to-end: architecture, security, automation, monitoring, and recovery. Always justify why others are not the best choice.

CHOOSE THE BEST AWS SOLUTION

5 DIFFICULT QUESTIONS + ANSWERS

EXAM FOCUS

- Evaluate trade-offs
- Compare AWS services
- Choose the BEST solution
- Operational excellence
- Cost, security, resilience



MOST OPERATIONALLY EFFICIENT



LEAST MANAGEMENT OVERHEAD



MOST SECURE



MOST RESILIENT



MOST COST-EFFECTIVE

1 MOST OPERATIONALLY EFFICIENT

A company runs a containerized web application with variable traffic. They want the solution that automatically scales and requires the LEAST operational effort.

- A. Amazon EC2 Auto Scaling
- B. Amazon ECS with EC2 launch type
- C. Amazon ECS with Fargate
- D. Amazon EKS with self-managed nodes

✓ ANSWER: C. AMAZON ECS WITH FARGATE

Fargate is serverless for containers, so AWS manages the underlying infrastructure, scaling, patching, and availability—minimizing operational overhead.

WHY OTHERS ARE WRONG:

- A: EC2 Auto Scaling requires you to manage servers.
- B: ECS with EC2 still requires server management.
- D: EKS needs cluster and node management.

2 LEAST MANAGEMENT OVERHEAD

You need a fully managed NoSQL database that automatically scales and requires minimal administrative tasks.

- A. Amazon RDS for MySQL
- B. Amazon DynamoDB
- C. Amazon Aurora Serverless v2
- D. Amazon ElastiCache for Redis

✓ ANSWER: B. AMAZON DYNAMODB

DynamoDB is fully managed, serverless, and automatically scales with no servers to manage.

WHY OTHERS ARE WRONG:

- A: RDS requires engine and instance management.
- C: Aurora still requires database management.
- D: ElastiCache requires cluster and node management.

3 MOST SECURE

You must store sensitive customer data with automatic encryption, fine-grained access control, and full audit logging.

- A. Amazon S3 with default encryption
- B. Amazon EFS
- C. Amazon S3 with SSE-KMS and bucket policies
- D. Amazon EBS with encryption

✓ ANSWER: C. AMAZON S3 WITH SSE-KMS AND BUCKET POLICIES

S3 with SSE-KMS provides server-side encryption using AWS KMS, bucket policies for access control, and CloudTrail for auditing.

WHY OTHERS ARE WRONG:

- A: Default encryption (SSE-S3) lacks fine-grained key control.
- B: EFS has weaker access controls and no object-level policies.
- D: EBS encryption does not provide fine-grained access policies.

4 MOST RESILIENT

You need a solution that keeps your application available even if an entire AWS Region fails.

- A. Multi-AZ deployment
- B. Multi-Region active-active deployment
- C. Auto Scaling across multiple AZs
- D. Backup to S3 in another Region

✓ ANSWER: B. MULTI-REGION ACTIVE-ACTIVE DEPLOYMENT

Active-active across multiple Regions keeps the application available through a full Region failure.

WHY OTHERS ARE WRONG:

- A: Multi-AZ protects against AZ failure, not Region failure.
- C: Auto Scaling across AZs still stays within one Region.
- D: Backups do not keep the app running.

5 MOST COST-EFFECTIVE

A dev team has a small app with low, unpredictable traffic. They want the solution that minimizes cost.

- A. EC2 always-on t3.small
- B. EC2 Auto Scaling group
- C. Lambda + API Gateway (Pay-per-request)
- D. ECS Fargate always running

✓ ANSWER: C. LAMBDA + API GATEWAY (PAY-PER-REQUEST)

Lambda charges only for actual execution time and API Gateway for requests, making it the most cost-effective for low and unpredictable traffic.

WHY OTHERS ARE WRONG:

- A: Always-on EC2 costs even when idle.
- B: Auto Scaling still keeps instances running.
- D: Fargate tasks incur cost while running.

SOLUTION EVALUATION FRAMEWORK



1. OPERATIONAL EFFICIENCY



2. MANAGEMENT OVERHEAD



3. SECURITY



4. RESILIENCE



5. COST EFFECTIVENESS



KEY TAKEAWAY

There is no single "best" service. The BEST solution depends on:

- Business requirements
- Trade-offs
- Risk tolerance
- Budget
- Operational capacity

TROUBLESHOOTING CHALLENGE

5 DIFFICULT QUESTIONS + ANSWERS

EXAM FOCUS

- Diagnose root cause
- Analyze logs & metrics
- Choose the BEST fix
- Avoid common traps
- Think like a DevOps pro



PIPELINES
DEPLOYMENTS



SERVERS
HEALTH CHECKS



PERMISSIONS
ACCESS



AUTOMATION
REMEDiation



MONITORING
LOGS & METRICS

1 PIPELINE SUCCEEDS BUT APPLICATION FAILS

Your CI/CD pipeline completes successfully, but users report errors after deployment. What is the MOST likely cause?

- A. Pipeline lacks rollback stage
- B. Application health check path is incorrect
- C. Build artifacts were not signed
- D. IAM role does not have S3 read access

✓ ANSWER: B. APPLICATION HEALTH CHECK PATH IS INCORRECT

If the health check endpoint is wrong or returns false positives, the pipeline will report success even though the app is unhealthy.

WHY OTHERS ARE WRONG:

- A: Rollback is for failures, not post-deploy health.
- C: Artifact signing does not affect runtime health.
- D: S3 access is needed during build, not after deployment.

2 DEPLOYMENT REPEATEDLY ROLLS BACK

Blue/Green deployment keeps rolling back within minutes after traffic shift. What should you check FIRST?

- A. CloudWatch alarm thresholds
- B. Target group health check configuration
- C. IAM permissions for CodeDeploy
- D. Auto Scaling group desired capacity

✓ ANSWER: B. TARGET GROUP HEALTH CHECK CONFIGURATION

Incorrect health check path, port, or thresholds cause targets to be marked unhealthy, triggering rollback.

WHY OTHERS ARE WRONG:

- A: Alarms may help, but health checks directly gate traffic.
- C: Permissions issue would stop deployment, not cause rollback.
- D: Capacity affects scaling, not rollback after traffic shift.

3 INSTANCES FAIL HEALTH CHECKS

New EC2 instances in an Auto Scaling group fail health checks and are terminated. What is the MOST likely cause?

- A. Security group blocks health check traffic
- B. AMI is not encrypted
- C. User data script logs are missing
- D. Instance type is unsupported

✓ ANSWER: A. SECURITY GROUP BLOCKS HEALTH CHECK TRAFFIC

If the load balancer cannot reach the instance on the health check port, it will mark it unhealthy and terminate it.

WHY OTHERS ARE WRONG:

- B: Encryption does not affect health checks.
- C: Logs help debugging but not the root cause.
- D: Unsupported type would fail launch, not fail health check.

4 CROSS-ACCOUNT DEPLOYMENT DENIED

CodePipeline in Account A fails to deploy to Account B with "Access Denied" error. What is the MOST likely issue?

- A. Missing trust relationship in Account B role
- B. Wrong S3 bucket encryption
- C. CodeDeploy agent not running
- D. Pipeline is in the wrong Region

✓ ANSWER: A. MISSING TRUST RELATIONSHIP IN ACCOUNT B ROLE

The role in Account B must trust the pipeline role from Account A using the correct external ID and permissions.

WHY OTHERS ARE WRONG:

- B: Encryption does not cause cross-account access denial.
- C: Agent issue happens after successful access.
- D: Region mismatch causes resource not found, not access denied.

5 AUTOMATED REMEDIATION DOES NOT TRIGGER

A CloudWatch alarm is in ALARM state, but the configured remediation (SSM Automation) does not run. What should you check FIRST?

- A. EventBridge rule pattern and target
- B. SSM document permissions
- C. Lambda function memory size
- D. Alarm evaluation periods

✓ ANSWER: A. EVENTBRIDGE RULE PATTERN AND TARGET

If the rule does not match the event or the target is misconfigured, the automation will not be triggered.

WHY OTHERS ARE WRONG:

- B: Document permissions matter only if execution starts.
- C: Lambda memory is irrelevant for SSM Automation.
- D: Alarm evaluation affects when alarm fires, not remediation setup.

TROUBLESHOOTING APPROACH (FOLLOW THIS ORDER)



1. IDENTIFY
THE SYMPTOM



2. CHECK LOGS
AND METRICS



3. FIND THE
ROOT CAUSE



4. APPLY THE
BEST FIX



5. VERIFY &
PREVENT RECURRENCE



KEY TAKEAWAY

Effective troubleshooting =
Right data, right order,
right solution.

- Validate assumptions
- Check configuration
- Use logs and metrics
- Fix the root cause

EXAM FOCUS

- Mixed real-world scenarios
- Multiple AWS domains
- Best answer selection
- Think like a DevOps pro
- 10 final challenge questions

FINAL MOCK EXAM

10 MIXED SCENARIO QUESTIONS



DEPLOYMENT



MONITORING



AUTOMATION



SECURITY



RESILIENCE

INCIDENT
RESPONSEINFRASTRUCTURE
AS CODE

1 A BLUE/GREEN DEPLOYMENT IN CODEDEPLOY COMPLETES SUCCESSFULLY, BUT USERS STILL HIT THE OLD VERSION. WHAT IS THE MOST LIKELY CAUSE?

- A. DNS TTL is too high
- B. Pipeline did not run
- C. Target group is unhealthy
- D. CodeDeploy failed to deploy

**ANSWER: A**

High DNS TTL delays traffic routing to the new environment.

2 CLOUDWATCH ALARM IS IN ALARM STATE, BUT THE ON-CALL ENGINEER RECEIVES NO NOTIFICATION. WHAT SHOULD YOU CHECK FIRST?

- A. SNS topic subscription
- B. Alarm threshold value
- C. CloudWatch agent status
- D. IAM role of the instance

**ANSWER: A**

If the SNS topic has no valid subscription, no notification is sent.

3 A SYSTEMS MANAGER AUTOMATION DOCUMENT RUNS SUCCESSFULLY BUT THE EXPECTED CHANGES DO NOT OCCUR ON THE INSTANCES. WHAT IS THE MOST LIKELY ISSUE?

- A. Incorrect target selection
- B. Automation document is invalid
- C. Maintenance window is closed
- D. Run Command timed out

**ANSWER: A**

The automation likely did not target the intended instances.

4 A LAMBDA FUNCTION CANNOT ACCESS SECRETS MANAGER AND RETURNS ACCESS DENIED. WHAT IS THE MOST LIKELY CAUSE?

- A. KMS key is disabled
- B. Lambda execution role lacks permission
- C. Secret is in a different Region
- D. Lambda timeout is too short

**ANSWER: B**

The execution role must have secretsmanager:GetSecretValue permission.

5 AN APPLICATION RUNNING IN MULTI-AZ EC2 AUTO SCALING GROUP BECOMES UNAVAILABLE WHEN ONE AZ FAILS. WHAT SHOULD YOU INVESTIGATE FIRST?

- A. Route 53 health checks
- B. ELB target group health checks
- C. AMI used for launch template
- D. CloudWatch log retention

**ANSWER: B**

If targets in the remaining AZ are unhealthy, traffic cannot be served.

6 YOUR TEAM NEEDS TO RESPOND TO A SECURITY INCIDENT AND IS INVESTIGATING API CALLS. WHAT IS THE BEST AWS SERVICE TO IDENTIFY WHO DID WHAT AND WHEN?

- A. AWS Config
- B. CloudTrail
- C. GuardDuty
- D. VPC Flow Logs

**ANSWER: B**

CloudTrail logs API activity with user, time, and source details.

7 A CLOUDFORMATION STACK UPDATE FAILS AND ROLLS BACK BECAUSE A NEW IAM ROLE ALREADY EXISTS. WHAT IS THE BEST WAY TO FIX THIS WITHOUT DELETING THE STACK?

- A. Import the existing resource
- B. Delete the stack and recreate it
- C. Disable rollback and retry
- D. Change the logical ID only

**ANSWER: A**

Import the existing resource into the stack and retry the update.

8 A CODEPIPELINE DEPLOYS SUCCESSFULLY, BUT USERS SEE HIGH ERROR RATE AFTER DEPLOYMENT. WHAT SHOULD YOU CHECK FIRST?

- A. Application logs and metrics
- B. Build logs in CodeBuild
- C. IAM permissions of the pipeline
- D. S3 artifact bucket policy

**ANSWER: A**

Post-deployment issues are usually application-level, not pipeline-level.

9 YOU NEED DR FOR AN APPLICATION WITH RTO OF 1 HOUR AND RPO OF 15 MINUTES AT THE LOWEST POSSIBLE COST. WHAT IS THE BEST STRATEGY?

- A. Multi-Region active-active
- B. Warm standby in another Region
- C. Pilot light in another Region
- D. Backup and restore only

**ANSWER: C**

Pilot light meets the RTO/RPO with lower cost than warm standby.

10 AN EVENTBRIDGE RULE DOES NOT TRIGGER THE LAMBDA FUNCTION AS EXPECTED. WHAT SHOULD YOU CHECK FIRST?

- A. Event pattern and sample event
- B. Lambda function memory size
- C. CloudWatch dashboard
- D. IAM role attached to Lambda

**ANSWER: A**

If the event pattern does not match the incoming event, the rule won't fire.

EXAM TOPICS COVERED



DEPLOYMENT



MONITORING



AUTOMATION



SECURITY



RESILIENCE

INCIDENT
RESPONSEINFRASTRUCTURE
AS CODE

SUCCESS REMINDER

- Read carefully
- Eliminate wrong answers
- Focus on best solution
- Think like a production engineer
- You've got this!

FINAL ANSWERS AND EXAM STRATEGY

ANSWERS TO PAGE 39 + STRATEGY TO PASS

ANSWERS TO PAGE 39 (FINAL MOCK EXAM)

- | | | |
|----|--|-----------|
| 1 | PIPELINE SUCCEEDS BUT APP FAILS | ANSWER: A |
| 2 | CLOUDWATCH ALARM NO NOTIFICATION | ANSWER: A |
| 3 | SYSTEMS MANAGER CHANGES NOT APPLIED | ANSWER: A |
| 4 | LAMBDA CANNOT ACCESS SECRETS MANAGER | ANSWER: B |
| 5 | MULTI-AZ ASG UNAVAILABLE AFTER AZ FAILURE | ANSWER: B |
| 6 | IDENTIFY WHO MADE API CALLS | ANSWER: B |
| 7 | CLOUDFORMATION UPDATE FAILS & ROLLS BACK | ANSWER: A |
| 8 | PIPELINE DEPLOYS BUT HIGH ERROR AFTER | ANSWER: A |
| 9 | DR STRATEGY (RTO 1 HOUR, RPO 15 MIN, LOW COST) | ANSWER: C |
| 10 | EVENTBRIDGE RULE DOES NOT TRIGGER LAMBDA | ANSWER: A |

ARCHITECTURE DECISION PATTERNS

- | | | |
|--|-----------------------------|---|
| | USE MANAGED SERVICES | Prefer services that remove infrastructure management (ECS, RDS, DynamoDB, etc.). |
| | DESIGN FOR FAILURE | Multi-AZ, auto scaling, retries, health checks, and graceful degradation. |
| | AUTOMATE EVERYTHING | Use pipelines, automation documents, IaC, and event-driven remediation. |
| | SECURITY BY DESIGN | Least privilege IAM, encryption, logging, monitoring, and compliance. |
| | OBSERVE AND IMPROVE | Metrics, logs, tracing, alarms, dashboards, learn from feedback. |

FINAL EXAM CHECKLIST

- | | |
|---------------------------------|---------------------------------------|
| ✓ Read the scenario carefully | ✓ Choose the BEST or LEAST |
| ✓ Identify the core requirement | ✓ Think operational excellence |
| ✓ Look for keywords | ✓ Consider cost, security, resilience |
| ✓ Eliminate wrong answers | ✓ Verify before selecting |
| ✓ Compare remaining options | ✓ Move on, manage your time |

EXAM STRATEGY FOCUS

- Understand WHY the answer is best
- Know WHY other options are wrong
- Recognize AWS keywords fast
- Use architecture decision patterns
- Follow a smart exam approach
- Stay calm, think like a DevOps pro

✓ WHY EACH CORRECT ANSWER WINS

- Best meets the requirement with the least risk.
- Aligns with AWS best practices and managed services.
- Provides the highest reliability, security, and scalability.
- Minimizes operational effort and human intervention.
- Delivers the required outcome with minimal cost.

✗ WHY DISTRACTORS ARE WRONG

- They are incomplete or require extra manual steps.
- They do not fully meet the requirement.
- They increase cost, complexity, or operational overhead.
- They introduce security or compliance gaps.
- They are not the MOST or LEAST as asked.

🔍 AWS KEYWORD RECOGNITION

KEYWORD	WHAT IT USUALLY MEANS
MOST / BEST	Look for fully managed, highly available, scalable solutions.
LEAST	Least management, least steps, least operational overhead.
AUTOMATICALLY	Use services with built-in automation or managed features.
MINIMIZE	Reduce manual work, errors, or operational tasks.
HIGHLY AVAILABLE	Multi-AZ, health checks, auto recovery, no single point of failure.
SECURE	Encryption, IAM least privilege, logging, monitoring.
COST-EFFECTIVE	Right-size, serverless, pay-per-use, avoid idle resources.
RTO / RPO	Design for recovery time and data loss requirements.
WITHOUT MANAGING SERVERS	Use serverless or fully managed services.
NEED TO IDENTIFY / TRACK	CloudTrail, Config, Logs, VPC Flow Logs.

🕒 LAST-MINUTE STRATEGY

- ✓ Review key AWS services and their strengths.
- ✓ Study common failure patterns and fixes.
- ✓ Focus on automation, security, and resilience.
- ✓ Practice reading scenarios under time pressure.
- ✓ Trust your first good answer, don't overthink.
- ✓ Manage time: ~1 minute per question.
- ✓ Stay calm, think like a production engineer.



**YOU'VE PREPARED. YOU'VE PRACTICED.
NOW GO PASS WITH CONFIDENCE!**